

Predictive and Prescriptive Analytics Applied to Energy Systems

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Operations Research

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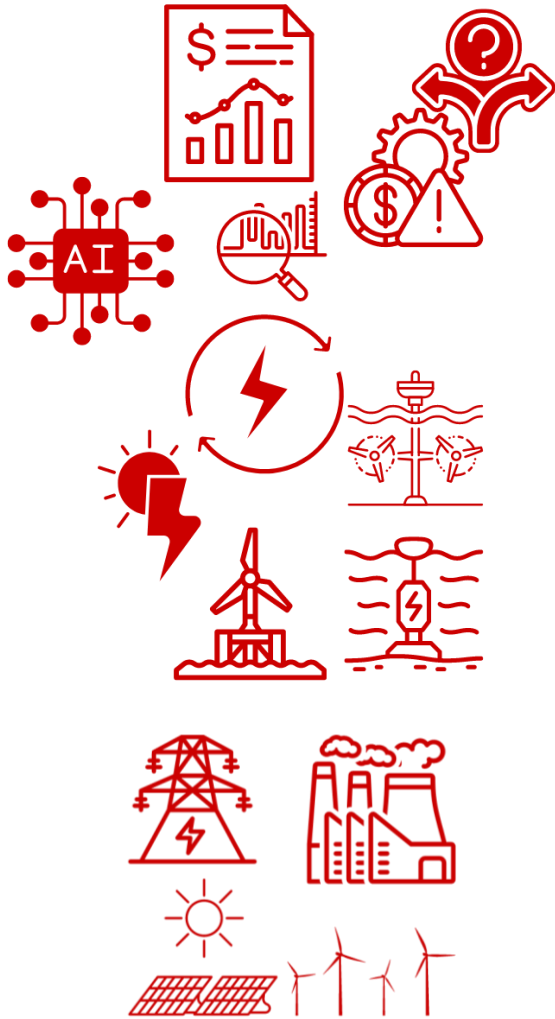
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Outline



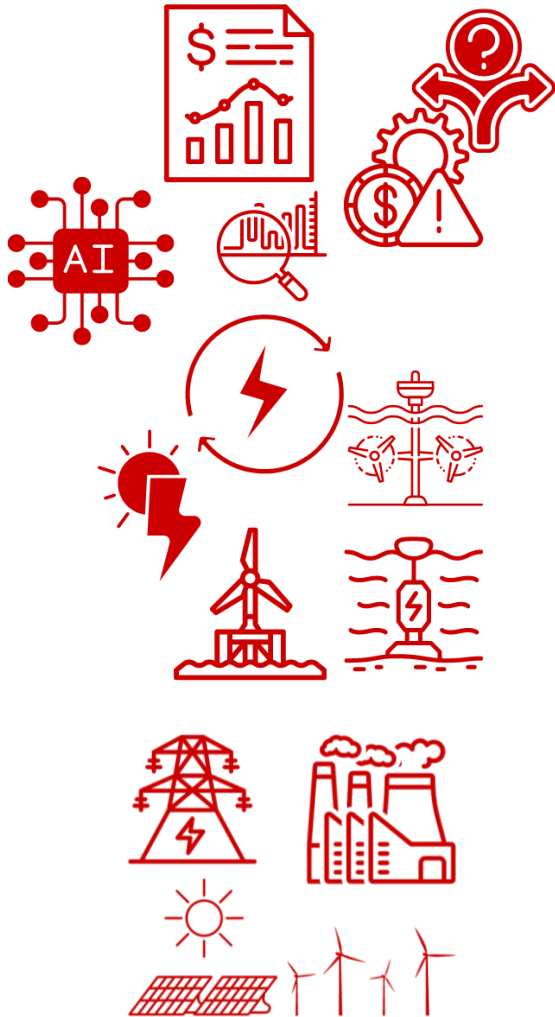
- Motivation
- Contributions:
 - Chapter 2: Optimizing Offshore Renewable Portfolios Under Resource Variability
 - Chapter 3: Scenario Generation and Risk-Averse Stochastic Portfolio Optimization Applied to Offshore Renewable Energy Technologies
 - Chapter 4: An Assessment of Multi-Layer Perceptron Networks for Streamflow Forecasting
 - Chapter 5: Capacity Expansion Planning on the North Carolina Energy System: Analyzing Policies, Cost Trajectories, and Hurricane Risks
- Closing Remarks

Definition



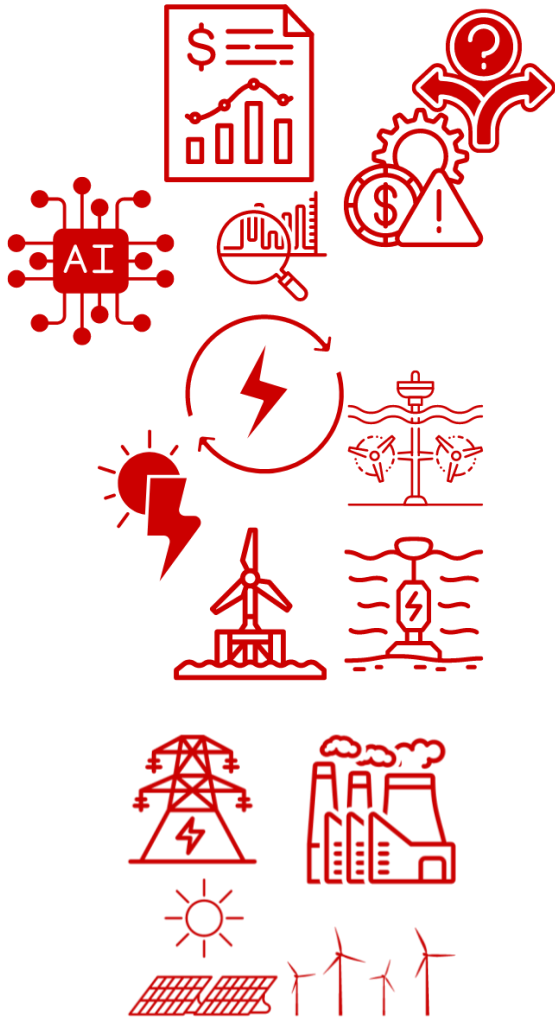
- Predictive analytics uses historical data, statistical algorithms, and machine learning techniques to estimate future outcomes
- Prediction of Electricity Prices, Load, **Streamflow**, Wind and Other Renewable Energy Resources
- Prescriptive analytics uses a combination of mathematical optimization, and statistical models to generate specific recommendations based on the estimates made by predictive analytics and/or existent data
- **Planning** and Operation

Motivation



- The power sector is a crucial aspect of the global economy, with **annual investments** surpassing **US\$2 trillion** worldwide
- Concerns regarding climate change and sustainability have fueled substantial investments in the renewable energy sector
- Since 2012 renewables have led the new power capacity additions in the world (86% during 2023 and 95% by 2026)
- It is expected that the annual global investments in the power sector would need to reach **US\$5 trillion by 2030** towards the target of the net-zero CO2 emissions by 2050

Motivation



- The higher penetration of renewable energy technologies poses additional challenges to the reliability and stability of electric power systems, exposing the sector to higher risk levels and costs
- **Concerns:** Variable nature of renewables; Evolution of Costs; Government Incentives; Uncertainty about extreme conditions (Low Energy Generation, Climate Change, Extreme Weather, etc.)
- Predictive and prescriptive models can help **reduce costs, improve the security of the power system and the assessment of risks**. Optimally allocating new capital, and helping to mitigate potential losses/failures that may jeopardize the decarbonization of the energy matrix

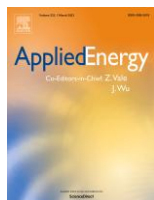
Chapter 1-2:

Offshore Portfolio Site Selection Models



Contributions:

- **Renewable Resource Integration:** Applied portfolio optimization for wind, wave, and ocean current resources off North Carolina; focused on balancing energy output with investment costs
- **Portfolio Optimization Advances:** Developed models using Mean-Variance and CVaR strategies, integrating LCOE and energy collection system considerations
- **Model Flexibility Enhancements:** Introduced model relaxation techniques to improve the computational efficiency of portfolio optimization models.
- **Innovation in Scenario Generation:** Utilized GAN models for scenario generation across large geographic areas; compared with Copula methods.



Optimizing Offshore Renewable Portfolios Under Resource Variability

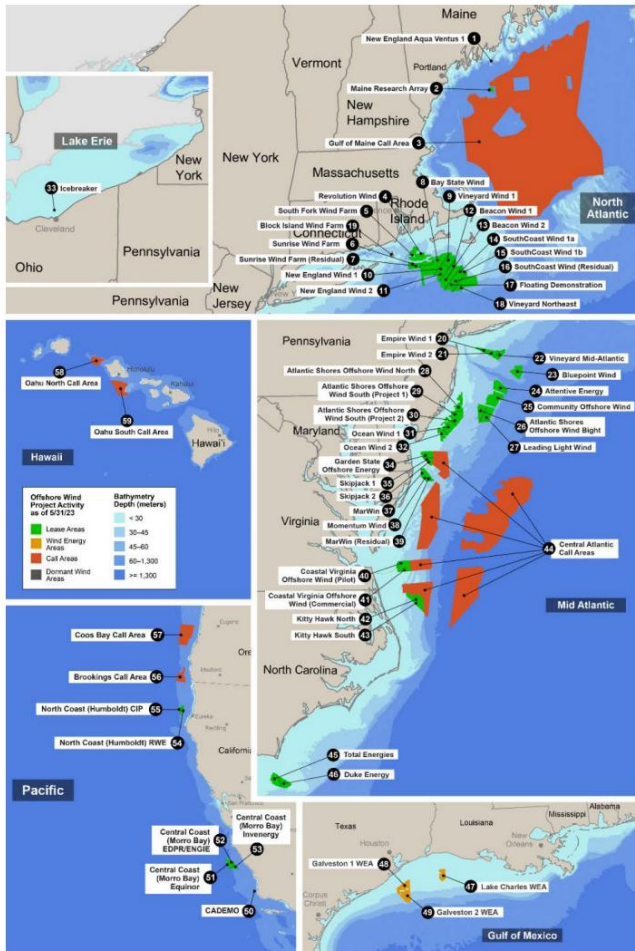


Scenario Generation and Risk-Averse Stochastic Portfolio Optimization Applied to Offshore Renewable Energy Technologies

de Faria, V. A., de Queiroz, A. R., and DeCarolis, J. F. (2022). Optimizing offshore renewable portfolios under resource variability. *Applied Energy*, 326. NREL (2021). Marine Energy in the United States: An Overview of Opportunities.

de Faria, V.A., de Queiroz, A. R., and DeCarolis, J. F. (2023). Scenario generation and risk-averse stochastic portfolio optimization applied to offshore renewable energy technologies. *Energy*, 325-120012.

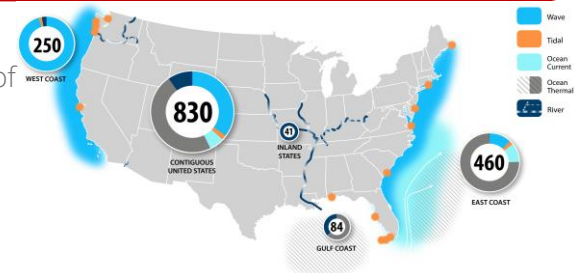
Locations of U.S. offshore wind energy pipeline activity and Call Areas (DOE, 2023)



- The US currently has around 17GW of offshore wind power offtake agreements.
- States Planning Targets and Mandates: 63GW by 2040 and 112GW by 2050
- In NC, Governor Cooper issued in 2021 the Executive Order No. 218 with targets for 2.8GW of offshore wind by 2030 and 8GW by 2040
- NC has other considerable offshore resources like wave and ocean currents

- What are the benefits of coordinating wind, wave, and ocean current resources in the North Carolina region, and how can this be optimally done?

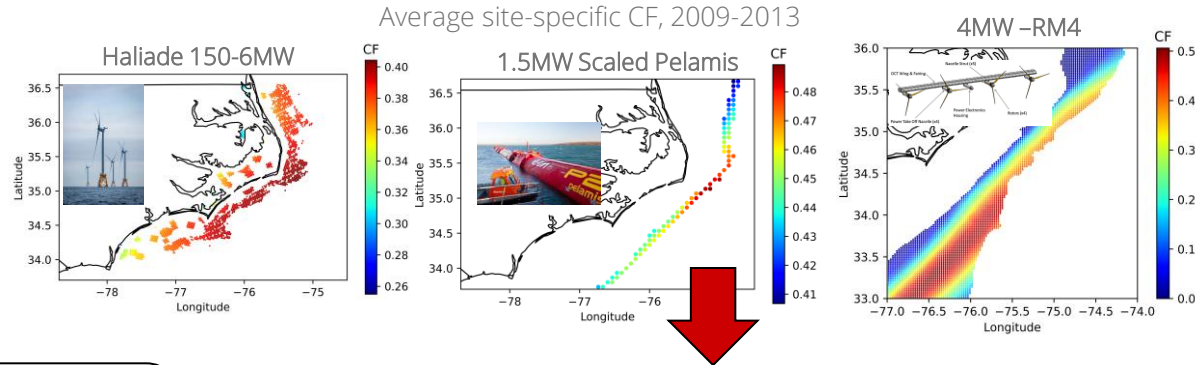
Technical Power Potential of
US Marine Energy
Resources [TWh/year].
(NREL, 2021)



Mean-Variance Portfolio Optimization (Model I)

Data

- Technology Type: Haliade 150-6MW **Wind** Turbine; Pelamis 1.5MW **Wave** Energy Device; RM4- 4MW **Ocean Current** Turbine
- Viable Site Locations For The Multiple Devices
- Site-Technology Cost
- Time Series of Energy Generation



Using this time series data, we can determine a variance-covariance matrix (Σ), which maps how each site location and technology correlates with each other

For a set of site candidates and technologies, we compute a time series of energy generation over a number of years, e.g., 2009-2013

$$\min Y^T \Sigma Y$$

(1) → Minimize Total Variability in Energy Generation

(2) → Bound the cost per MWh

(3) → Total # turbines per tech

(4) → Max # turbines per site & tech

(5) → Limit the size of the energy collection system

(6) →

Variables

$y_{e,i} \in Y$: Number of turbines deployed of the technology $e \in E$ at the site $i \in I_e$, $y_{e,i} \in \mathbb{Z}^+$

$v_{e,i}$: Binary variable responsible for controlling the center of the energy collection system for each energy technology, $v_{e,i} \in \{0,1\}$

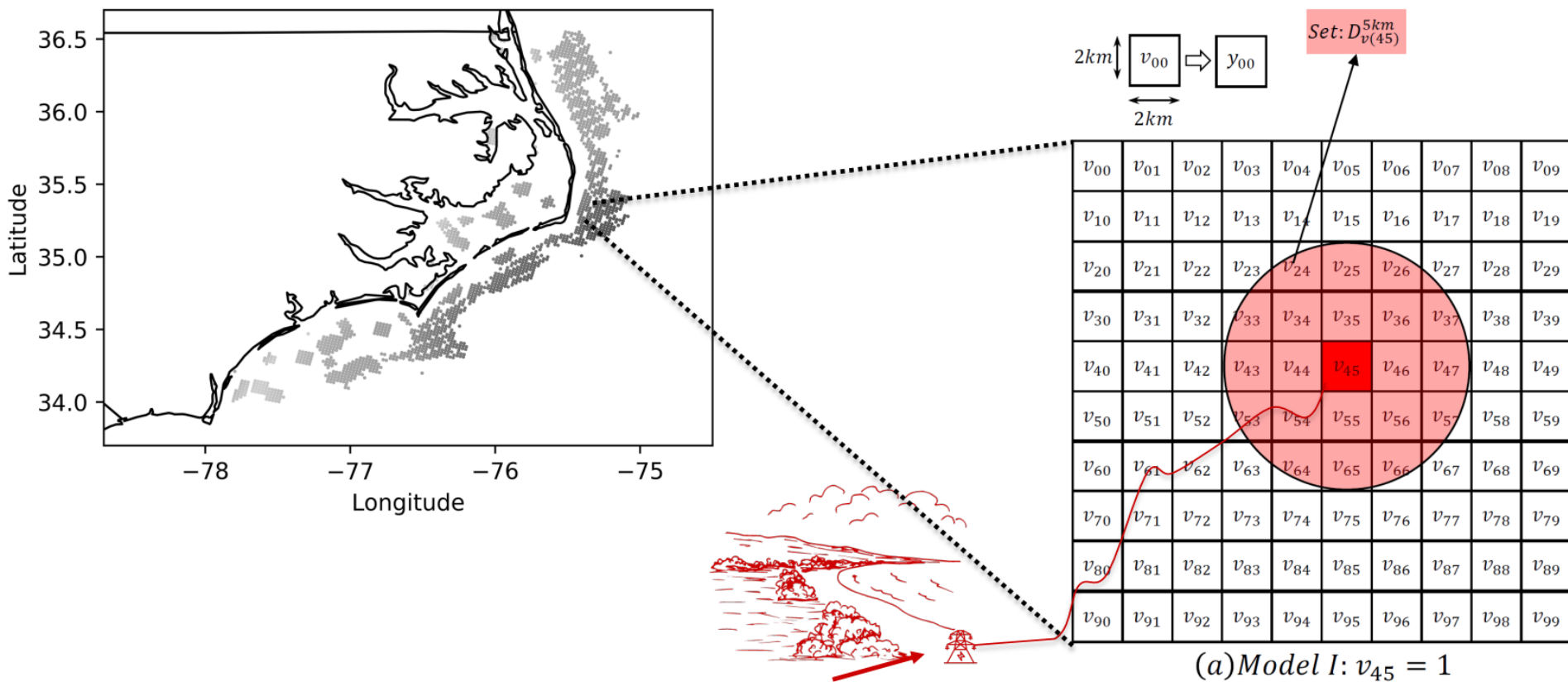
$$\frac{\sum_{e \in E} \sum_{i \in I_e} \overset{\text{Cost } [\$/]}{C_{e,i}} y_{e,i}}{\sum_{e \in E} \sum_{i \in I_e} \overset{\text{Avg. Gen [MWh]}}{EG_{e,i}} y_{e,i}} \leq \overline{LCOE} \text{ [\$ / MWh]}$$

$$\sum_{i \in I_e} y_{e,i} = TNt_e \quad \forall e \in E$$

$$y_{e,i} \leq \overline{Nt_{e,i}} \quad \forall (e,i) \in (E, I_e)$$

$$\sum_{k \in D_{v(e,i)}^R} y_{e,k} \geq v_{e,i} TNt_e \quad \forall (e,i) \in (E, I_e)$$

$$\sum_{i \in I_e} v_{e,i} = 1 \quad \forall e \in E$$



Portfolio Optimization

The Need For a Modeling Relaxation (Model II)

- Problem:** Simulating Model I for a large coastal area can lead to prohibitive computational times
- Model Relaxation:** Simplify the constraints or objective function of a mathematical optimization problem, providing an upper/lower bound on the optimal value of the original problem

$$\min X^T \Sigma X \quad (1)$$

s. t.

$$\frac{\sum_{e \in E} \sum_{i \in I_e} C_{e,i} x_{e,i}}{\sum_{e \in E} \sum_{i \in I_e} EG_{e,i} x_{e,i}} \leq \overline{LCOE} \quad (2)$$

$$\sum_{i \in I_e} x_{e,i} = TNt_e \quad \forall e \in E \quad (3)$$

$$x_{e,i} \leq \overline{Nt_{e,i}} \quad \forall (e,i) \in (E, I_e) \quad (4)$$

$$\sum_{k \in D_{w(e,j)}^R} x_{e,k} \geq w_{e,j} TNt_e \quad \forall (e,j) \in (E, J_e) \quad (5)$$

$$\sum_{j \in J_e} w_{e,j} = 1 \quad \forall e \in E \quad (6)$$

Variables Model I

$y_{e,i} \in Y$: Number of turbines deployed of the technology $e \in E$ at the site $i \in I_e$, $y_{e,i} \in \mathbb{Z}^+$

$v_{e,i}$: Binary variable responsible for controlling the center of the energy collection system for each energy technology, $v_{e,i} \in \{0,1\}$

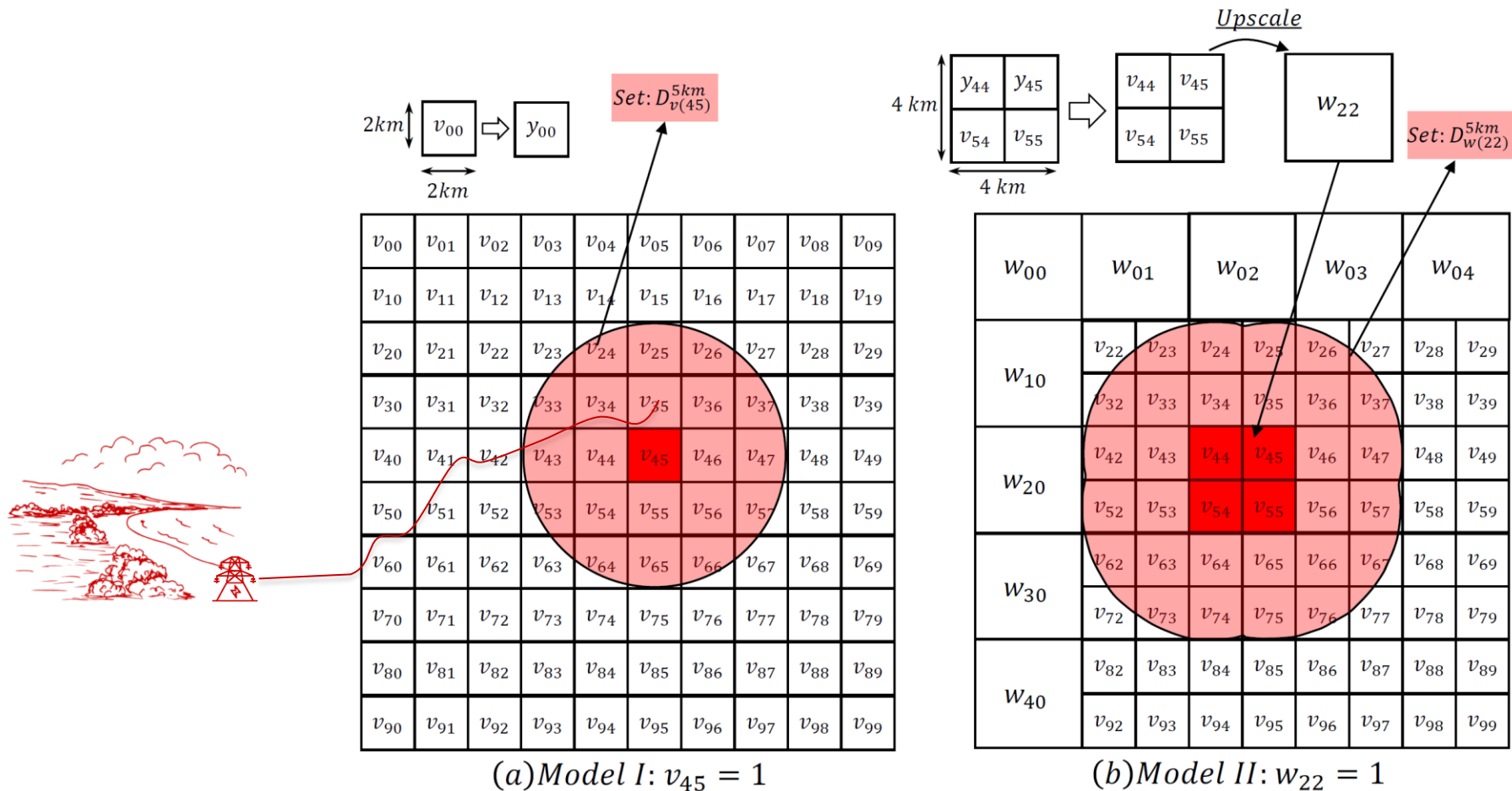


Variables Model II

$X_{e,i} \in X$: A relaxation for the number of turbines deployed of the technology $e \in E$ at the site $i \in I_e$, $x_{e,i} \in \mathbb{R}^+$

$w_{e,j}$: A relaxation for the binary variable responsible for controlling the center of the energy collection system for each energy technology, $w_{e,j} \in \{0,1\}$. Upscaled version of $v_{e,i}$

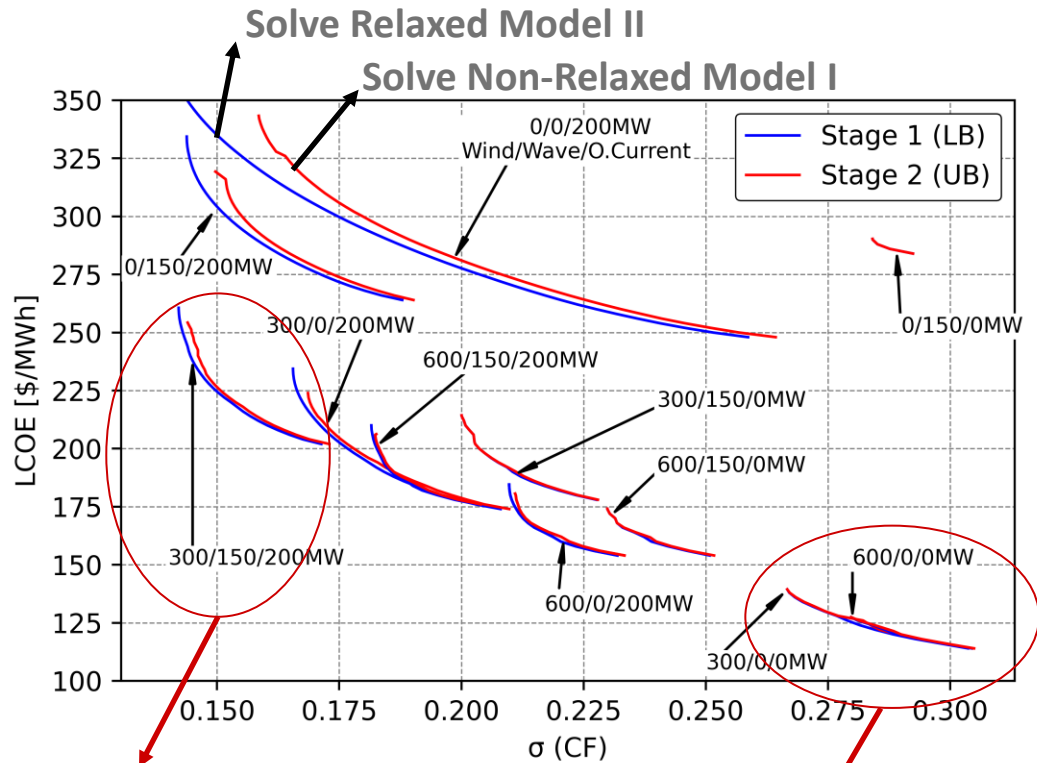
Portfolio Optimization ($v_{e,i}$ to $w_{e,j}$)



Model relaxation of the site locations. Nearby sites are aggregated to reduce the number of integer variables

Efficient Frontiers

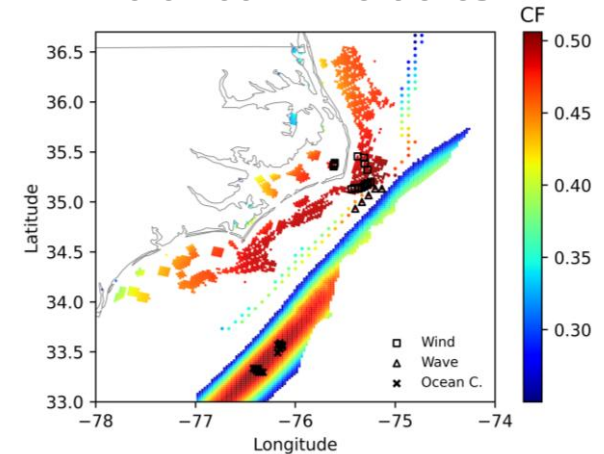
Efficient Frontier For Different Deployments of Wind, Wave, and Ocean Current



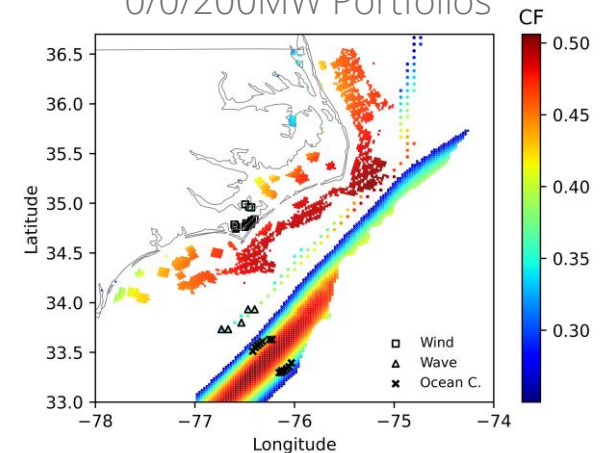
Wind + Wave + O.Current: Low Energy Variability but High LCOE

Only Wind: Low LCOE but High Energy Variability

Optimal Site Locations For The Lowest LCOE in the 300/0/0, 0/150/0, and 0/0/200MW Portfolios



Optimal Site Locations For The Lowest σ (CF) in the 300/0/0, 0/150/0, and 0/0/200MW Portfolios



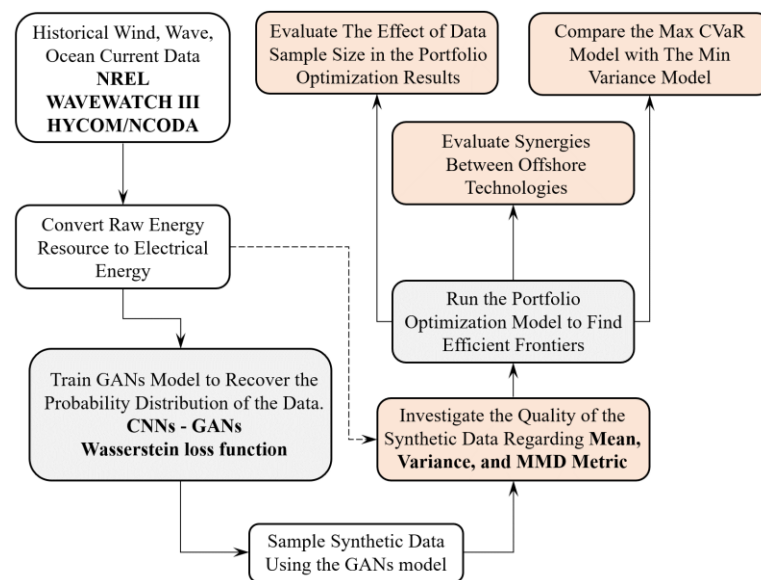
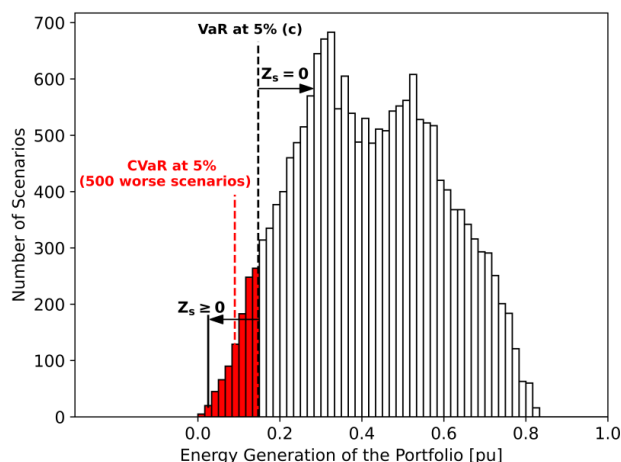
Scenario generation and risk-averse stochastic portfolio optimization applied to offshore renewable energy technologies.

What is missing in the previous model we presented



- Variance is a symmetric function → **penalize conditions of high and low generation**
- Another important point to consider in the portfolio optimization analysis is the **data availability**; with several energy resources, matching data is hard!

CVaR → expected Generation of the portfolio at the 5% worse scenarios (red)



de Faria, V. A., de Queiroz, A. R., and DeCarolis, J. F. (2023). Scenario generation and risk-averse stochastic portfolio optimization applied to offshore renewable energy technologies. *Energy*, 325-120012.

Risk-Averse Portfolio Optimization

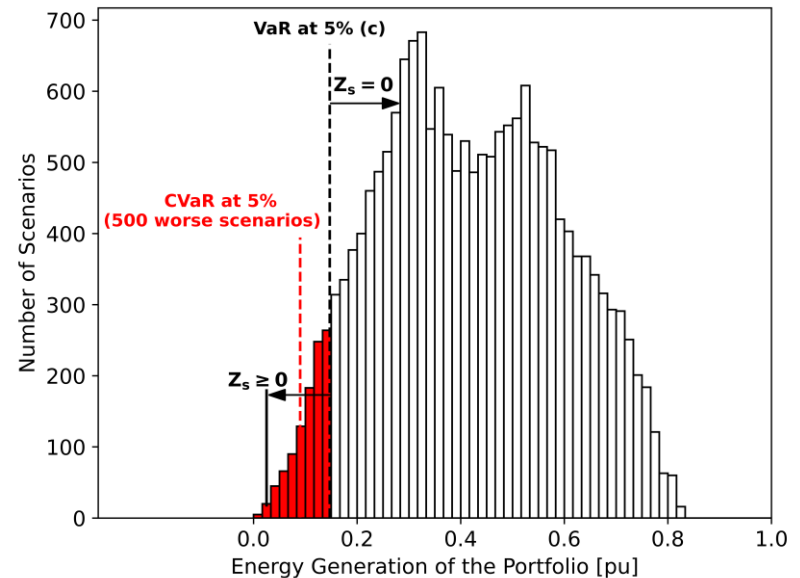
$$\begin{aligned}
 & \max_{(x,v)} CVaR_\alpha(X,\zeta) & (1) \\
 \text{s.t. } & \frac{\sum_{e \in E} \sum_{i \in I_e} C_{e,i} x_{e,i}}{\sum_{e \in E} \sum_{i \in I_e} \mathbb{E}[EG_{e,i,\zeta}] x_{e,i}} \leq \overline{LCOE} & (2) \\
 & \sum_{i \in I_e} x_{e,i} = TNt_e \quad \forall e \in E & (3) \\
 & x_{e,i} \leq \overline{Nt_{e,i}} \quad \forall (e,i) \in (E, I_e) & (4) \\
 & \sum_{k \in D_{e,i}^R} x_{e,k} \geq v_{e,i} TNt_e \quad \forall (e,i) \in (E, I_e) & (5) \\
 & \sum_{i \in I_e} v_{e,i} = 1 \quad \forall e \in E & (6) \\
 & x_{e,i} \in \mathbb{Z}^+ \quad \forall (e,i) \in (E, I_e) & (7) \\
 & v_{e,i} \in \{0,1\} \quad \forall (e,i) \in (E, I_e) & (8)
 \end{aligned}$$

$$\max_{(x,v,c,z)} c - \frac{1}{\alpha N_s} \sum_{s=1}^{N_s} z_s \quad (11)$$

$$z_s \geq c - \sum_{e \in E} \sum_{i \in I_e} EG_{e,i,\zeta_s} x_{e,i} \quad \forall s \in \{1, \dots, N_s\} \quad (12)$$

$$c \geq 0 \text{ and } z_s \geq 0 \quad (13)$$

- A Limitation of the Mean-Variance Model is that it Penalizes Both Sides of the Probability Distribution (Low and High Energy Generation)
- Can We Be More Precise in The Region We Want to Optimize? **Let's Condition Our Risk to Low Energy Generation Scenarios!**



Synthetic Data Generation (GANs and Copulas)

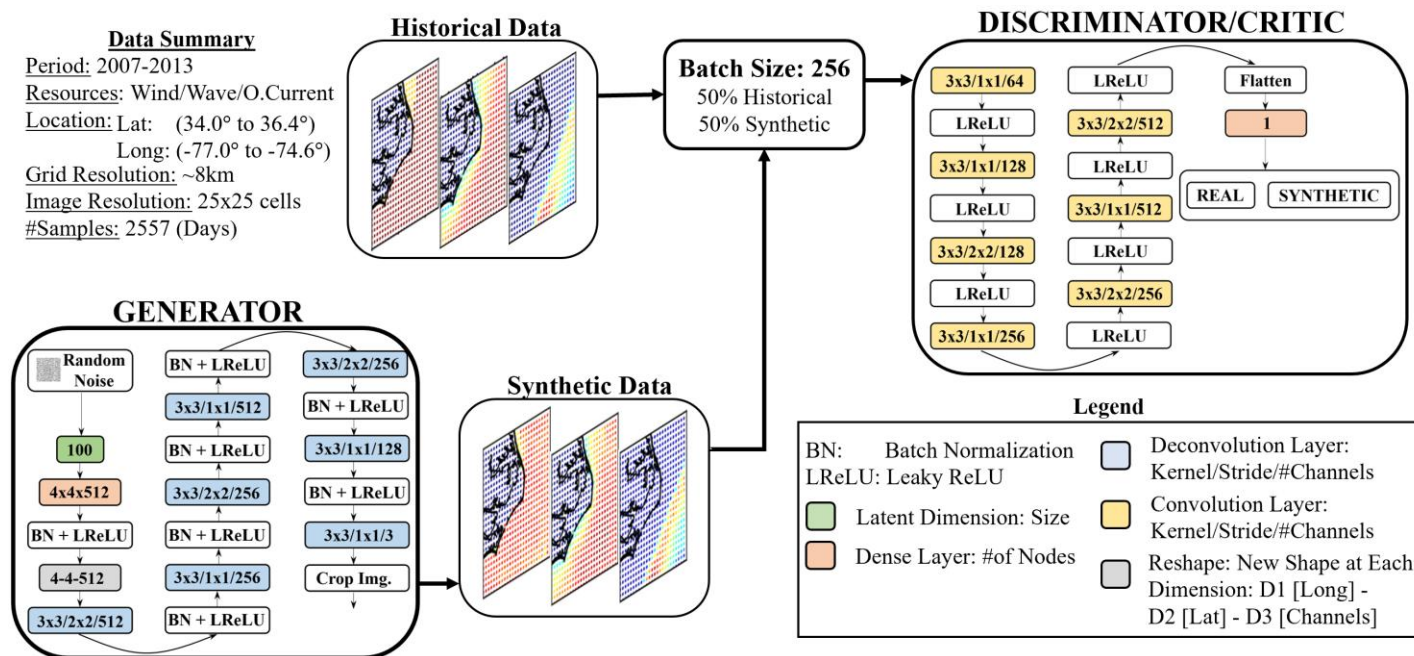


Fig. Generative Adversarial Neural Network Model Diagram (W-GANS)

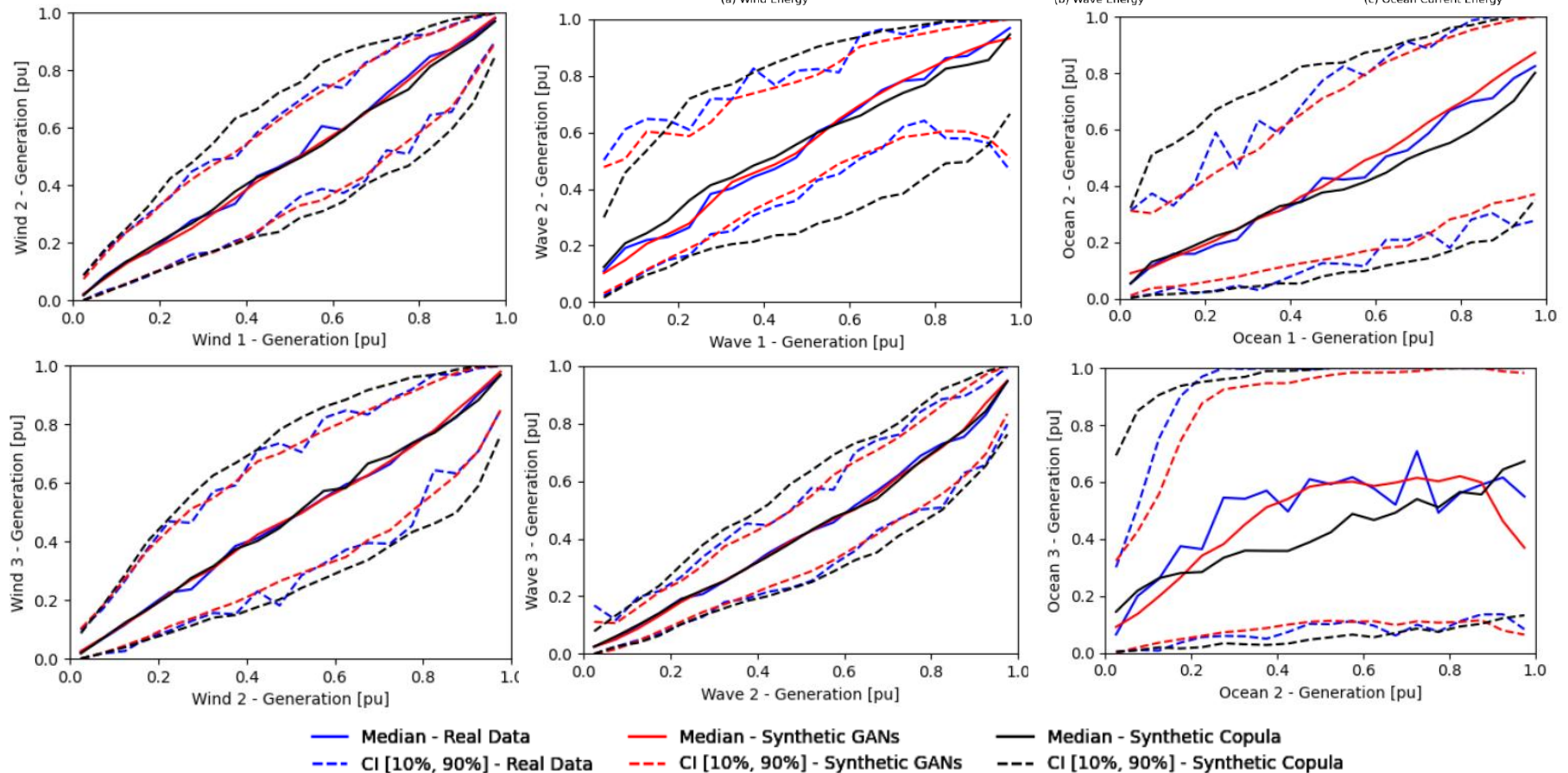
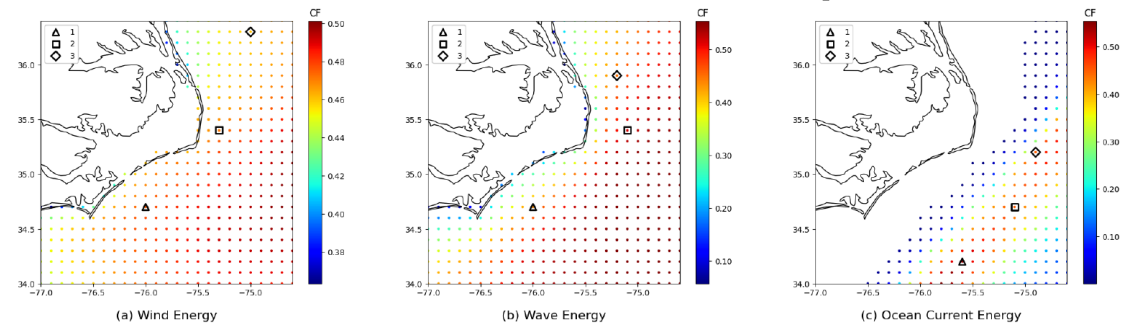
Copulas:

- Preserve the marginal distributions of each site location and build the joint probability distribution with some of the properties of the original data like the correlation between sites (Variance-Covariance Matrix) – Gaussian Copula

$$C(u_1, \dots, u_n, R) = \Phi_G(\Phi^{-1}(u_1), \dots, \Phi^{-1}(u_n); R);$$

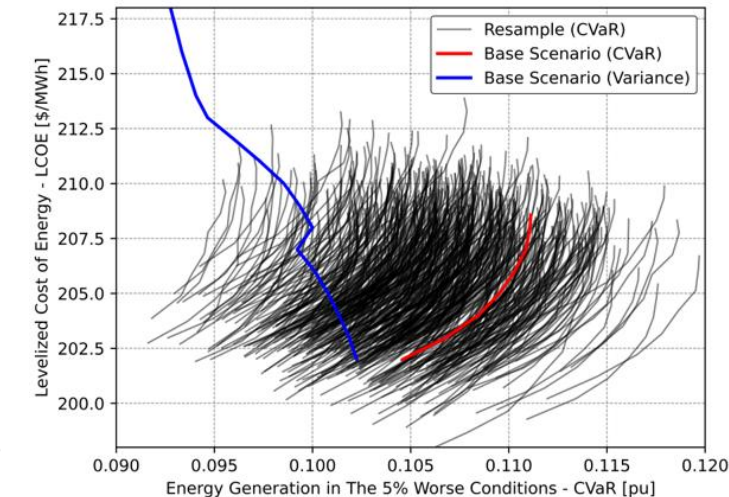
Sample Statistics Hist. Data, GANs, Copulas

Probability Distribution of the **Historic** (Blue) **GANs** (Red) and **Copula** (Black) Data:
Pairwise Comparison of How Different Technologies/Sites Interact With One Another

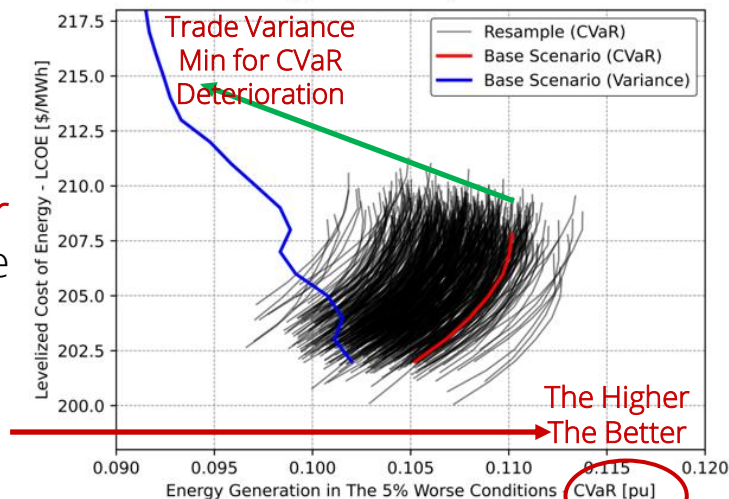


Portfolio Optimization Results

Efficient Frontier
Given the
Deployment of
200 Wind
(1.2GW), 200
Wave (0.3GW),
and 200 Ocean
Current Turbines
(0.8GW),
Considering
Uncertainty
Quantification
and the Use of
Different Number
of Samples in The
Portfolio
Optimization (10
and 30 Years)

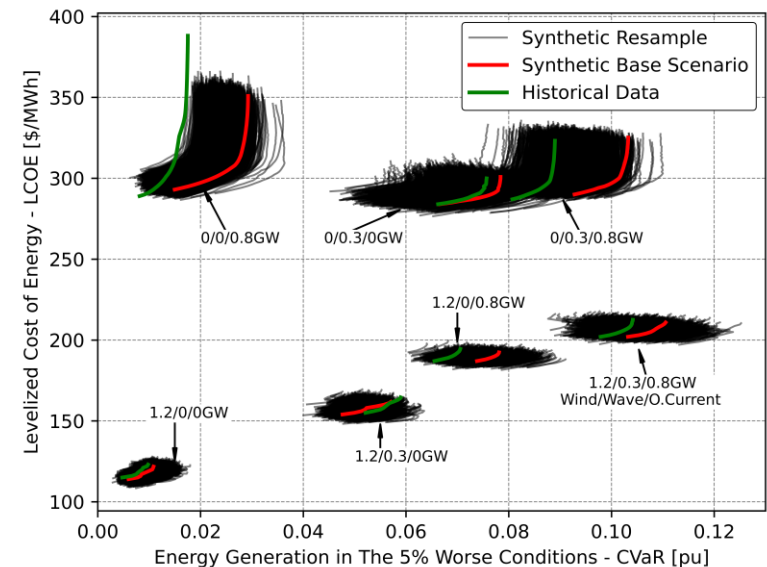


(a): 10 Year Sample



(b): 30 Year Sample

Efficient Frontier for Different
Combinations of Wind, Wave and
Ocean Current Considering Seven
Years of Historical and Synthetic
Data. Only The min-CVaR Model Was
Used in These Simulations



Chapter 3:



An Assessment of Multi-Layer Perceptron Networks for Streamflow Forecasting



Contributions:

- **Multilayer Perceptron Networks for Streamflow Forecasting:** Assessed the effectiveness of multilayer perceptron networks in forecasting streamflow for large-scale hydropower in Brazil
- **Comparative Analysis:** Evaluated the performance of multilayer perceptron against the streamflow forecasting models currently employed by the Brazilian Independent System Operator (ISO)
- **Optimization of MPL Configurations**

Motivation

- In the US, hydropower accounts for almost 30% of the total renewable energy generation (254 TWh - 2022)
- In Brazil +70% of the TOTAL electricity generated comes from hydro (441 TWh -2022)
- In hydro-dominant systems like Brazil the marginal price of electricity (\$/MWh) is directly affected by the system storage level in the reservoirs
- Streamflow forecasting plays a critical role in defining the optimal system operation and electricity prices, especially in the **short-term energy market**

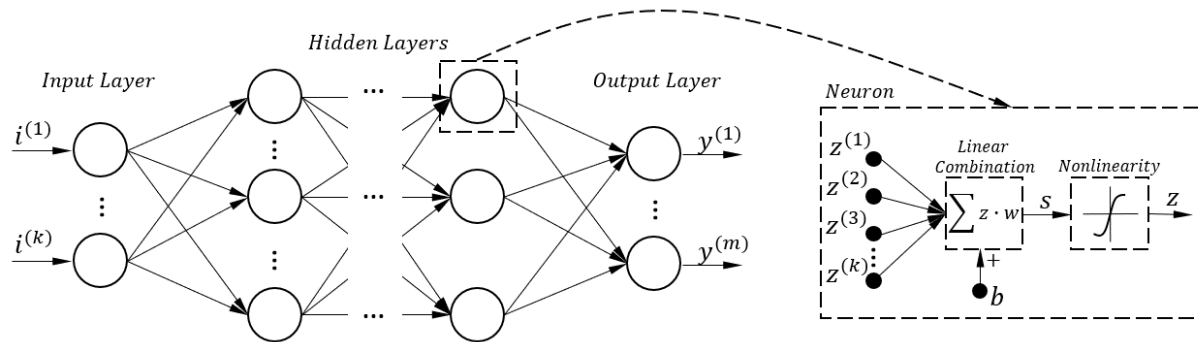


Objectives:

- Compare the performance of MLPs with the models used by the Brazilian ISO (Independent System Operator)
- Investigate different optimization algorithms, objective functions, activation functions, regularizations, and initialization strategies

MLP Configurations

- Forecast Natural Streamflow **14 days ahead** in daily time discretization.
- Inputs: Historical streamflow and rainfall (01/2000 ~ 12/2016)



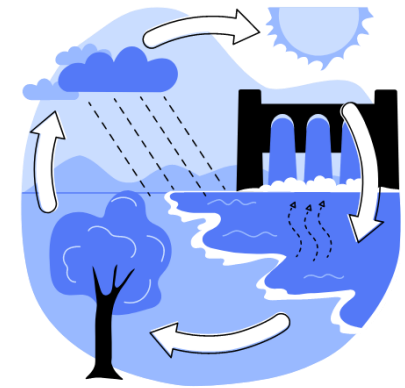
- Optimization Algorithms: GDM, RMSProp, Adam
- Objective Functions: $MSE, MAPE^2$
- Regularization Strategies: Dropout, ℓ_2
- Activation Functions: Sigmoid, Tanh, ReLU
- Parameter Initialization: $N(0,1)$, With Variance control (Xavier)

Error Metrics

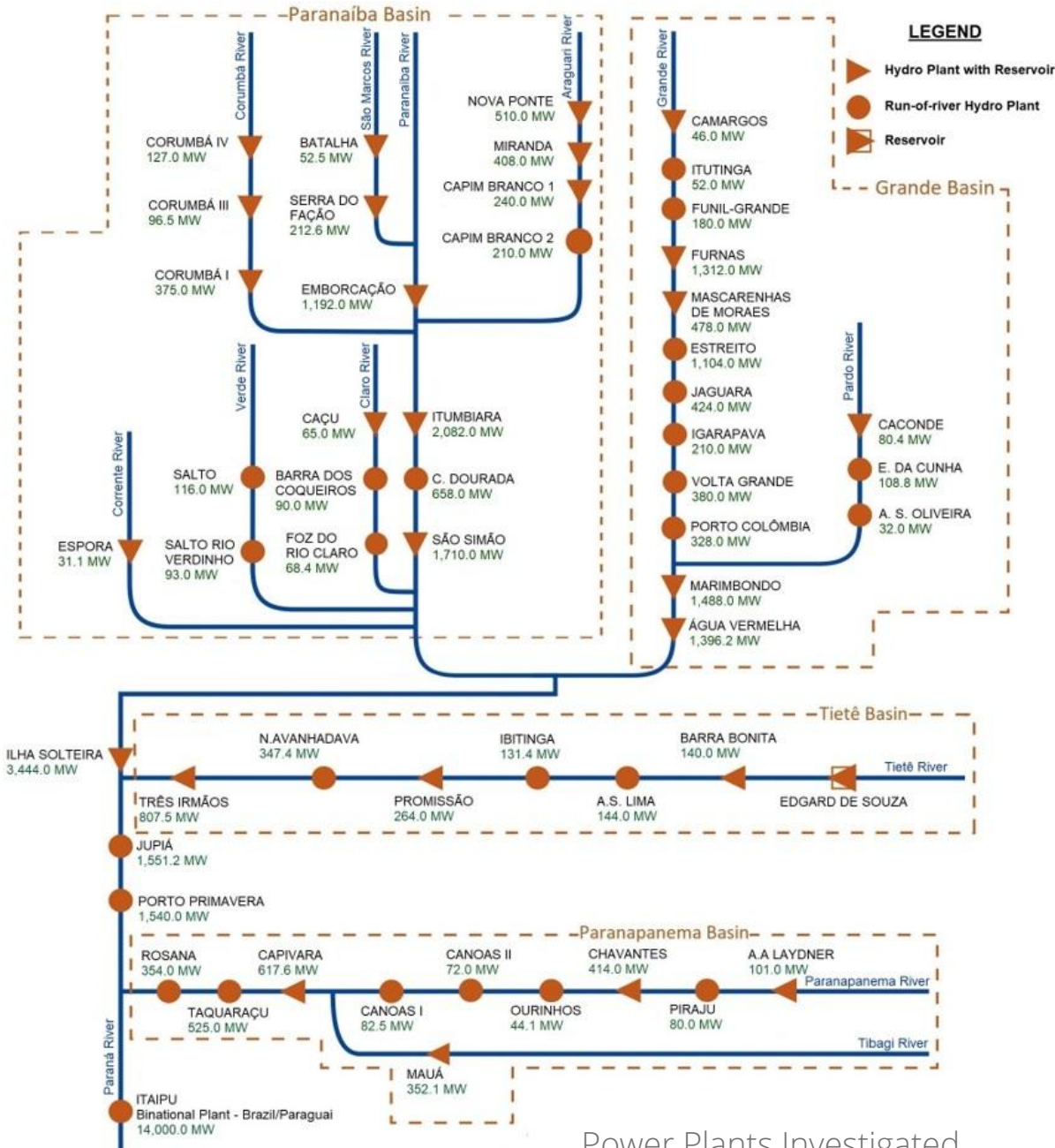
$$MAPE_a = \frac{100}{N_s} \sum_{i=1}^{N_s} \left| \frac{\overline{y^{(i)}} - \overline{\hat{y}^{(i)}}}{\overline{y^{(i)}}} \right| \%$$

$$NSE = 1 - \frac{\sum_{i=1}^{N_s} \left(\overline{y^{(i)}} - \overline{\hat{y}^{(i)}} \right)^2}{\sum_{i=1}^{N_s} \left(\overline{\hat{y}^{(i)}} - \mu_0 \right)^2}$$

Brazilian ISO Model



- Models Officially Used by the ISO:
 - SMAP (Soil Moisture Accounting Procedure):
 - Rainfall-runoff model
 - Simplified representation of the water cycle with four equivalent reservoirs
 - It uses a combination of physical, mathematical, and conceptual elements to describe the movement of water from precipitation to evaporation and runoff, and the interactions between these components
 - Rainfall from ECMWF and ETA (maintained by the ISO)
 - Previvaz: AR, ARMA, PAR, PARMA
 - Do not use rainfall
- From these different models the ISO selects for each hydro power plant the alternative with the best accuracy

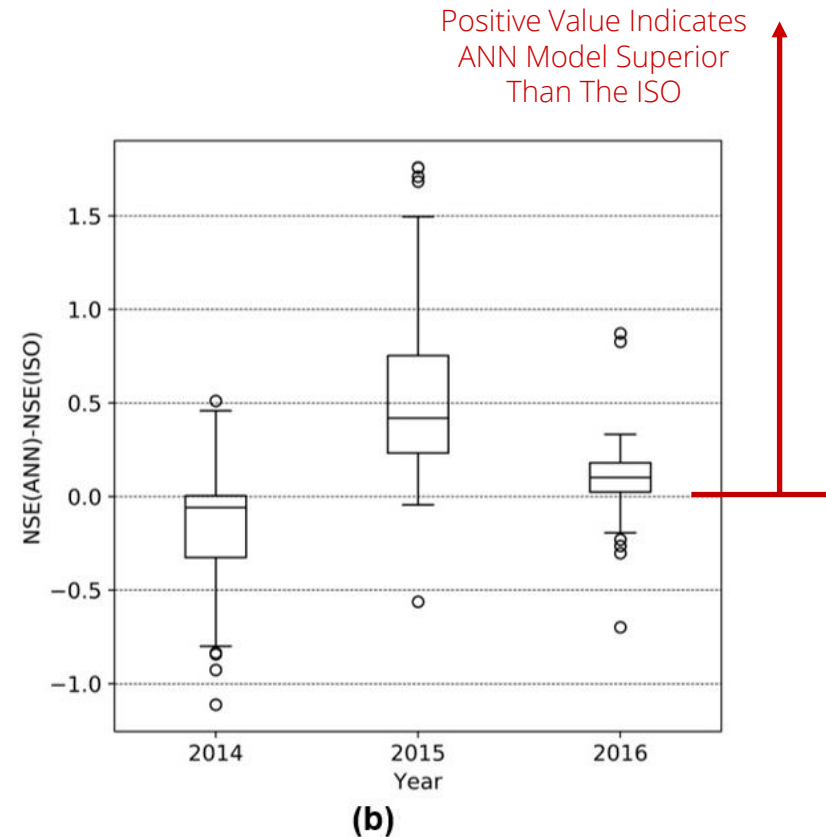
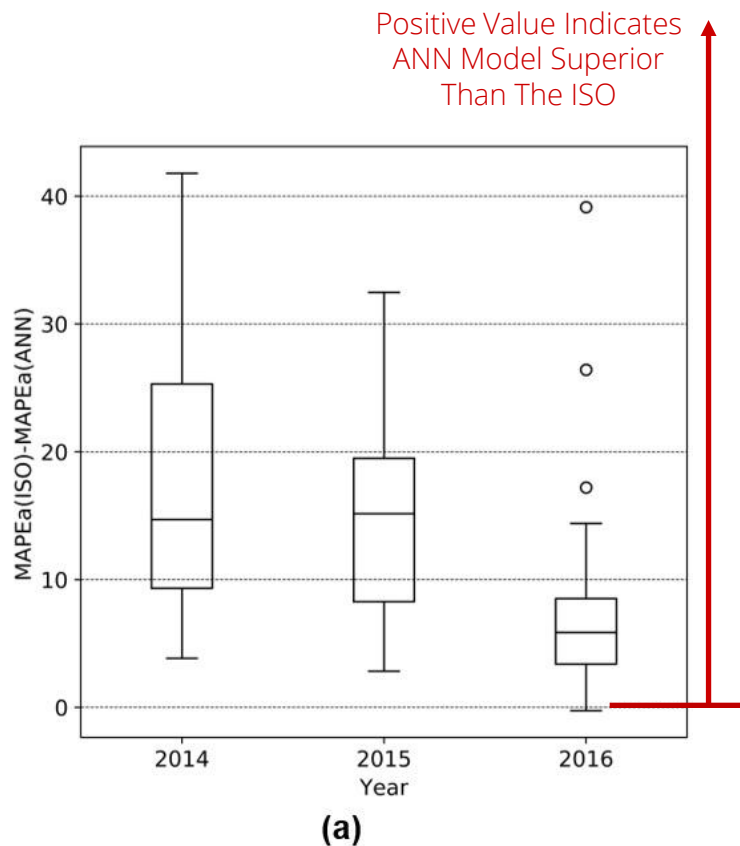


A total of 55 hydropower plants were investigated located in the Paraná Basin. Which accounts for more than 30% of the total electricity generated in the country



Power Plants Investigated.

Comparison of the Forecasting Error Between the Brazilian ISO Models and ANNs from 2014 to 2016 (Test Set) - Boxplot MAPE (a) and NSE (b)

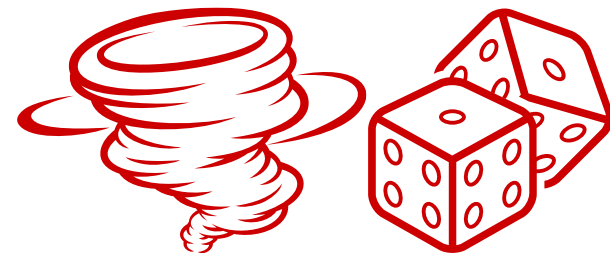
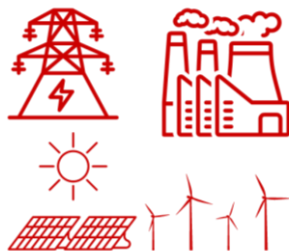


Error Metrics

$$MAPE_a = \frac{100}{N_s} \sum_{i=1}^{N_s} \left| \frac{\overline{y^{(i)}} - \overline{\hat{y}^{(i)}}}{\overline{y^{(i)}}} \right| \%$$

$$NSE = 1 - \frac{\sum_{i=1}^{N_s} \left(\overline{y^{(i)}} - \overline{\hat{y}^{(i)}} \right)^2}{\sum_{i=1}^{N_s} \left(\overline{\hat{y}^{(i)}} - \mu_0 \right)^2}$$

Chapter 4:



Capacity Expansion Planning on the North Carolina Energy System: Analyzing Policies, Cost Trajectories, and Hurricane Risks

Contributions:

- **Policy and Technology Analysis:** Examined alternative government policies, technological trajectories, and fuel price scenarios impacting capacity expansion in North Carolina's power sector
- **Hurricane Risk Assessment:** Utilized a multistage stochastic capacity optimization model to analyze the risk of hurricane damage to North Carolina's system infrastructure.
- **Constructed a Database for the North Carolina Energy System**
- **Infrastructure Damage Computation Method**
- **Fragility Curves Database Development**

Long-term Energy Systems Planning

Capacity Expansion Models

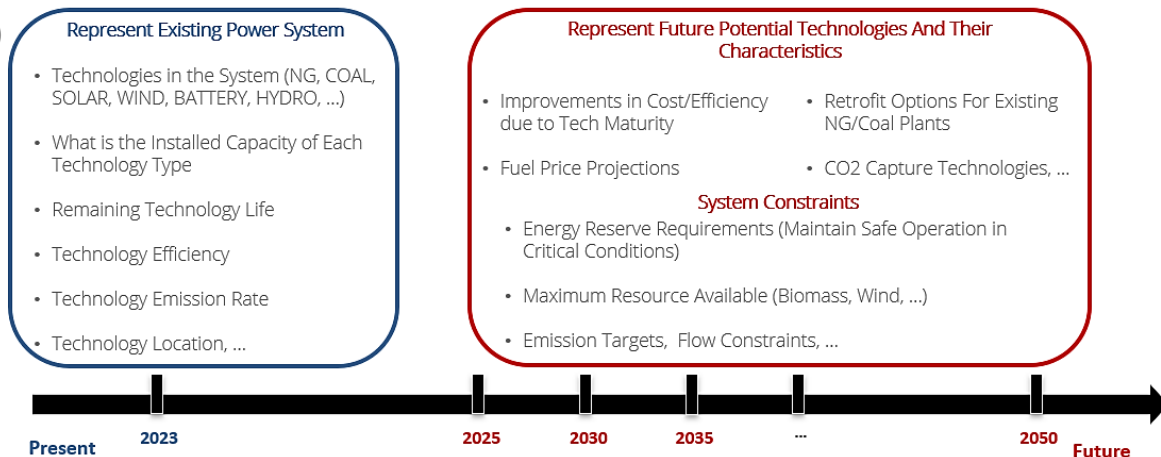
Optimize and simulate **energy systems** for a pre-determined future horizon to **decide when, how much, place, and type of investments to be made**

Consider **physical characteristics, technical** and **interactions among system elements**, energy policy, and markets

$$\begin{aligned} \min & cx \\ \text{s.t. } & Ax = b \\ & x \geq 0 \end{aligned}$$

We have many inputs to represent:

- Existing Generators Capacity (EIA-860, EIA)
- Emissions (NREL-ATB, EIA-906/920, EIA)
- Efficiencies (EIA-906/920, NREL-ATB)
- Fuel Prices (EIA-AEO)
- Technology Prices (NREL-ATB)
- Improvements in CFs (NREL-ATB)
- Max Capacity (NREL Models, Utilities)
- Max Activity (ORNL-Billion Ton Report)
- Energy Imports (EIA)
- Energy Reserve Requirements/Capacity Credits (Utilities, NERC)
- Technology Life (NREL, EIA-860)
- Government Policies/CO2 Targets



Capacity Expansion Formulation

Model Simplifications:

- Choose a finite number of decision years (e.g., 2023, 2025, 2030, ..., 2050)
- Discretize each decision year in a small number of seasons (e.g., Winter, Spring, Summer, Autumn)
- Discretize each season in a limited number of representative days (e.g. 1,2,3 days)
- Discretize each day in a limited number of representative time steps (e.g. 24h, 8h, 4h)
- Choose a Limited Number of Regions
- Aggregate technologies...

Minimize: TotalCost
= LoanCost + FixedCost + VariableCost

**Objective
Function**

$$ACT_{p,s,d,t,v,r} = \sum_{(i,o)} FO_{p,s,d,t,v,r,i,o} \\ \forall \{p,s,d,t,v,r\} \in \Theta_{act}$$

Process activity

$$(CF_{s,d,t,v,r} \cdot C2A_t \cdot SEG_{s,d} \cdot TLF_{p,t,v,r}) \cdot CAP_{t,v,r} \geq ACT_{p,s,d,t,v,r} \quad \forall \{p,s,d,t,v,r\} \in \Theta_{act}$$

**Technology
Capacity**

$$\sum_{(i,t,v)} FO_{p,s,d,t,v,r,i,c} = DEM_{p,c,r} \cdot DSD_{s,d,c,r} \\ \forall \{p,s,d,r,c \in C^d\} \in \Theta_{demand}$$

**Supply-
demand**

$$FO_{p,s,d,t,v,r,i,o} = EFF_{t,v,i,o} \cdot FI_{p,s,d,t,v,r,i,o} \\ \forall \{p,s,d,t \in T - T^{st}, v,r,o\} \in \Theta_{flow}$$

**Process-Level
Commodity
Flow Constraint**

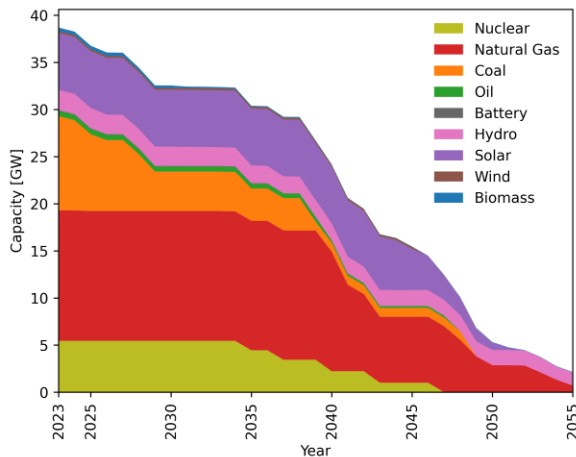
$$\sum_{(s,d,t,v,i,o)} (EAC_{t,v,r,i,o,e} \cdot FO_{p,s,d,t,v,r,i,o}) \leq ELM_{p,e,r} \\ \forall \{p,r,e \in C^e\} \in \Theta_{demand}$$

**Emission limit
constraint**

Other Constraints and Bounds: Capacity/Activity Limits, Storage, Life of Technology, Retrofit Options, Planning Reserve Margin (Capacity Credit), Curtailment, Energy Imp/Exp

The North Carolina Energy System

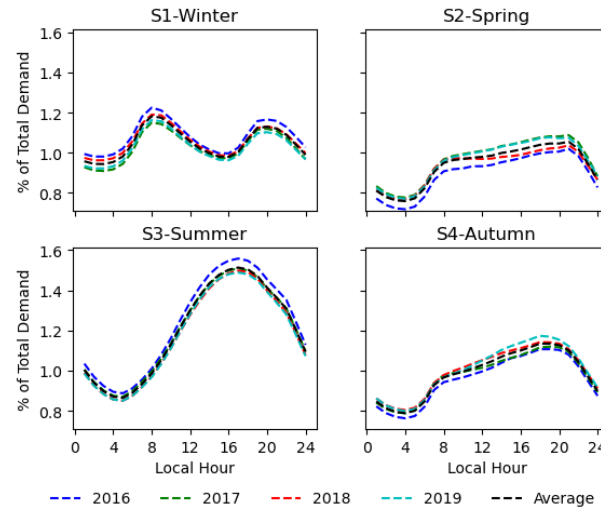
Fig 1. Total Capacity for Existing Technologies. Capacity Decreases as Existing Technologies Retire



- About 36% Capacity From NG
- 25% From Coal (Many Retirements Soon)
- 15% Solar
- 14% Nuclear
- 5% Hydro
- ~5% (Biomass, Land-Wind, ...)

>85% of Total Electricity Generated by NG, Coal and Nuclear

Fig 2. Load Distribution on NC



On NC the critical load usually happens during the early winter mornings. Leading to the enforcement of a 17% energy reserve requirement.

- Solar energy has a very low contribution to meeting this reserve (Low Capacity Credit)

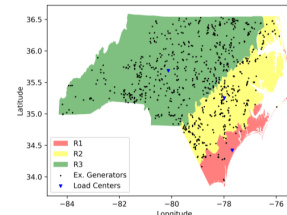


Fig 3. CF Distribution For Utility and Distributed PV

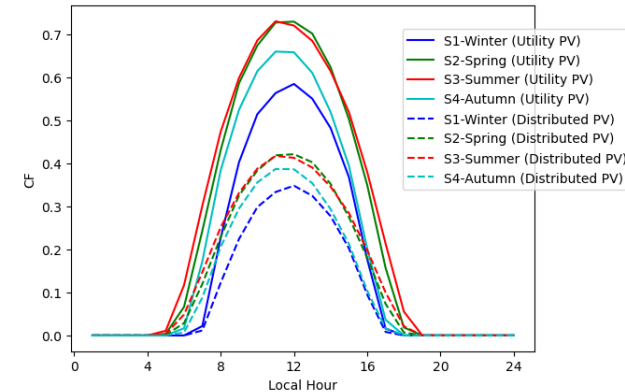
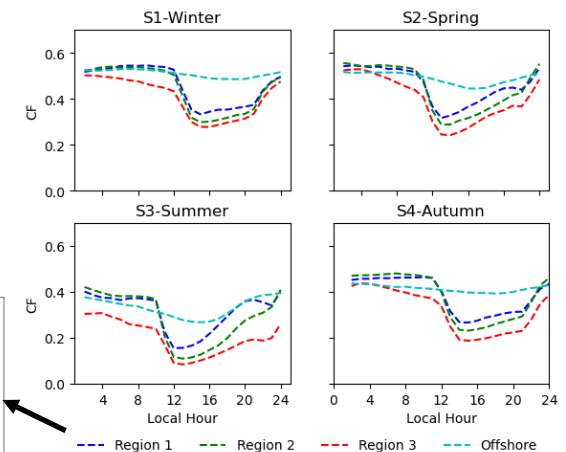


Fig 4. CF Distribution For Wind Technologies



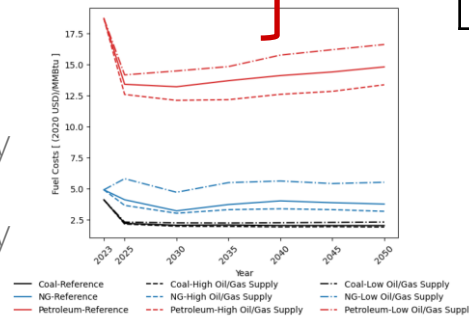
Simulation Scenarios

Policies:

- Emission Limits** (House Bill 951, 2021):
 - 70% CO₂ reduction from 2005 levels by 2030
 - Carbon neutrality by 2050
- Min Capacity of Offshore Wind** (EO. No. 218, 2021)
 - 2.8 GW by 2030
 - 8 GW by 2040

Fuel Price Trajectories (EIA-AEO):

- Low Prices: High Oil and Gas Supply
- Reference: Reference Conditions
- High Prices: Low Oil and Gas Supply



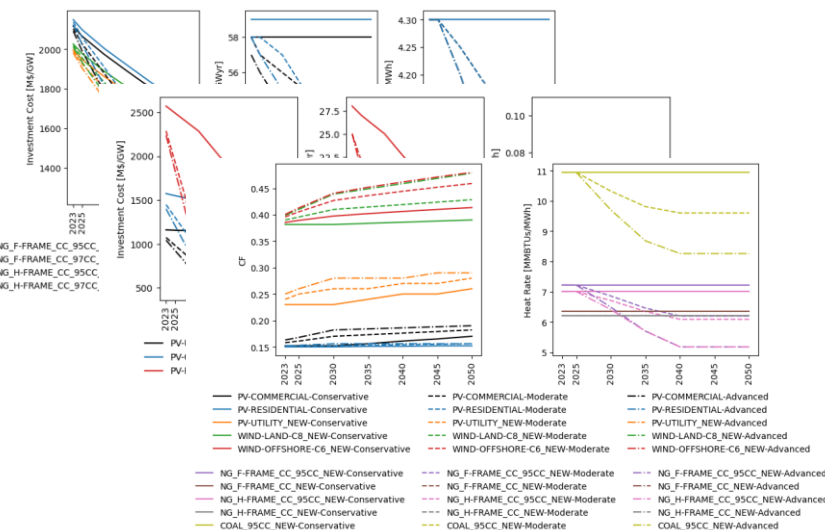
Consider Three Policy Scenarios:

Policy Scenario	CO ₂ Targets	Offshore Wind Targets
CO ₂ -OffWind	Yes	Yes
CO ₂ -NO_OffWind	Yes	No
NO_CO ₂ -NO_OffWind	No	No

Technology Trajectories (NREL-ATB): These trajectories affect not only the technology cost but also efficiency, capacity factors (renewables) and heat rates (thermal power plants)

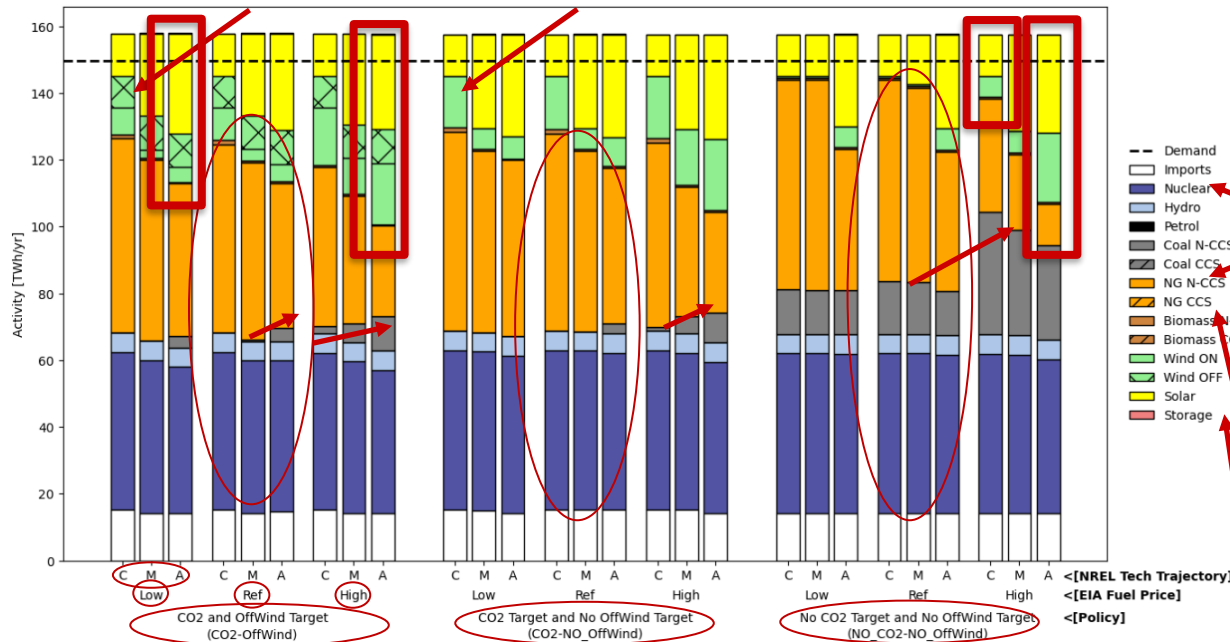


- Conservative (C): Conservative Technology Innovation
- Moderate (M): Moderate Technology Innovation
- Advanced (A): Advanced Technology Innovation



3 Policies x 3 Fuel Paths x 3 Technology Paths =
27 Scenarios

Activity [TWh/yr]

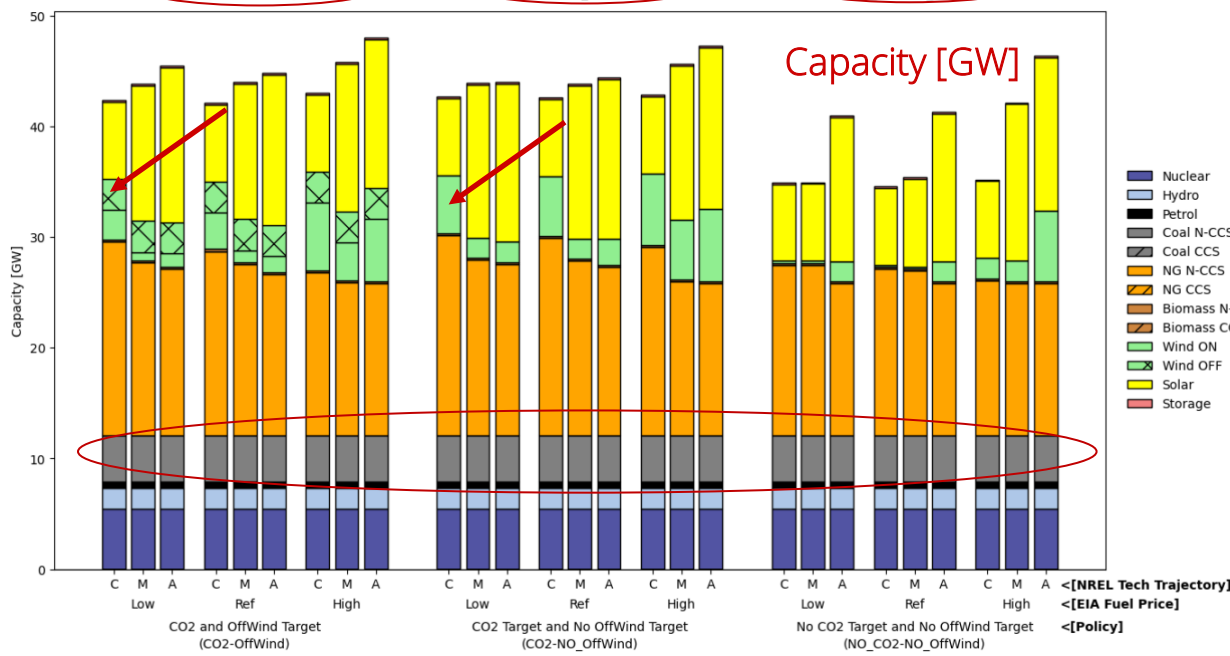


- No New Coal Capacity (But a significant amount not retired yet). This capacity gets utilized in scenarios of higher renewable participation and high fuel prices

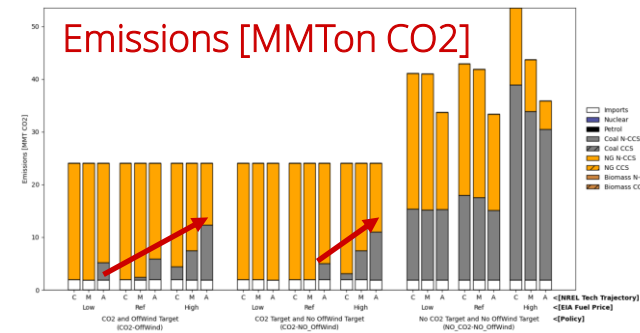
Nuclear and NG continue to be the primary providers of electricity by 2030

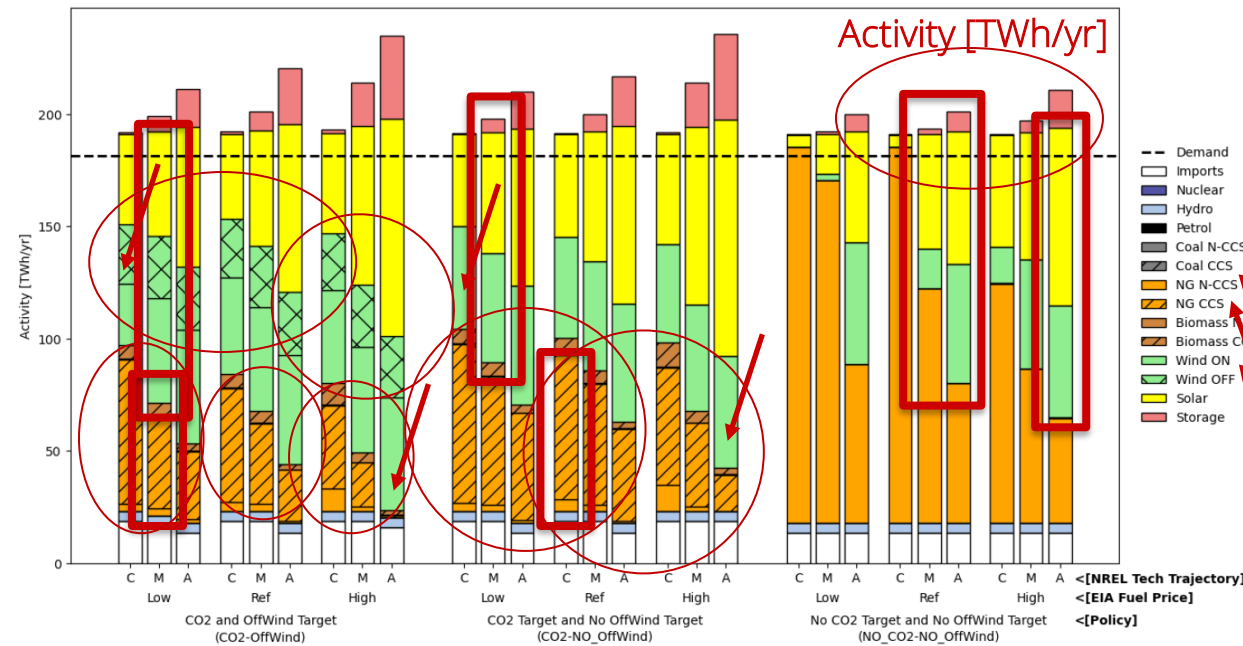
- Renewable participation increases significantly under Advanced tech paths and high fuel prices taking space from NG.
- Offshore wind does not get deployed naturally in the system by 2030. When it is enforced it can take space from onshore wind
- The model does not use CCS, and finds deploying renewables more economical
- No significant participation of storage

Capacity [GW]



Emissions [MMTon CO2]





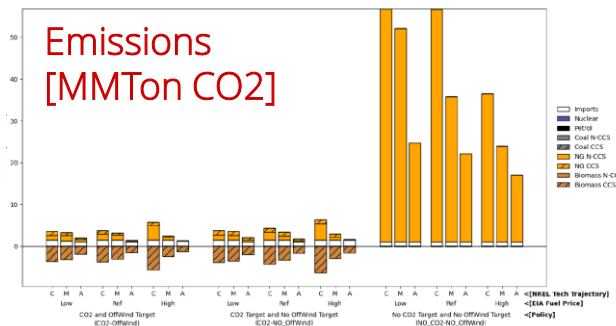
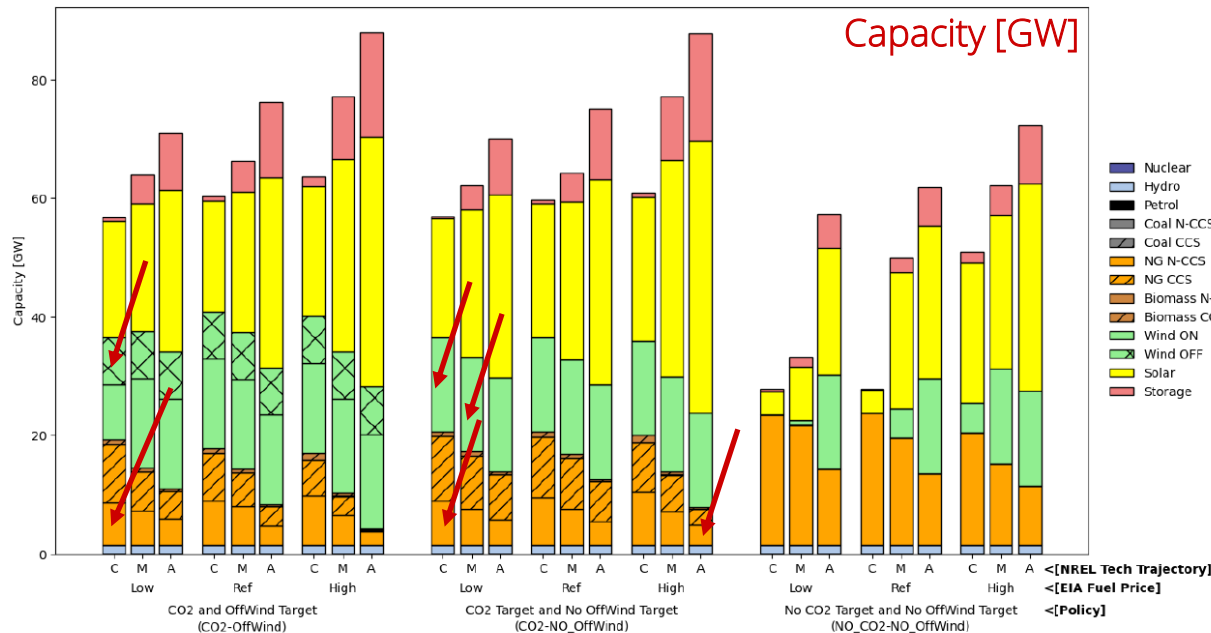
In **CO2-limited** scenarios, renewables become the major providers of electricity by 2050. However, **NG-CCS** plays a significant role

Even without CO2 targets, **renewable** energy still plays a **significant** role in many scenarios. For scenarios under high fuel prices, renewables can become the primary providers of electricity

In scenarios with CO2 targets, **NG-CCS** and **Biomass-CCS** are very prevalent in the model solution. With NG generally being used to meet reserve requirements and **peak load conditions**

Even in scenarios without CO2 targets it is possible to see a significant participation of **storage** technologies

Offshore wind does not get naturally deployed in the system by 2050, and when it is enforced, it can take space from NG and onshore wind



Total System Cost (Discounted to 2022 USD) Divided by Total Electricity Demand

CO2 Target	Off Wind Target	Fuel Trajectory (EIA AEO)	Electricity Cost [\$/MWh]		
			Technology Trajectory (NREL ATB)		
			Conservative	Moderate	Advanced
Y	Y	Low	30.8	29.0	27.7
Y	Y	Ref	31.3	29.4	28.1
Y	Y	High	32.6	30.6	28.9
Y	N	Low	27.4	26.3	25.2
Y	N	Ref	28.0	26.8	25.6
Y	N	High	29.4	28.1	26.6
N	N	Low	25.4	25.2	24.7
N	N	Ref	26.2	25.9	25.2
N	N	High	28.3	27.4	26.3

Annotations:

+21% (from 30.8 to 32.6)

+7% (from 27.4 to 29.4)

-8% (from 31.3 to 29.4)

-4% (from 30.6 to 28.1)

-10% (from 28.0 to 26.8)

-4% (from 26.8 to 26.3)

+8% (from 25.2 to 26.6)

+4% (from 25.9 to 26.3)

 $\frac{25.19 - 24.74}{24.74} = 1.8\%$

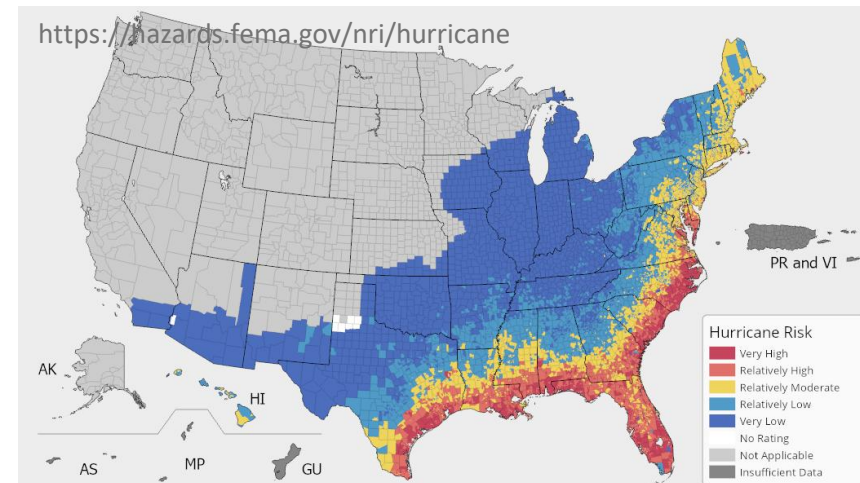
 $\frac{26.59 - 26.32}{26.32} = 1\%$

- If technologies evolve following an **Advanced NREL-ATB path**, the difference in cost between operating the system enforcing or not CO2 targets is between **1% (High fuel prices) to 1.8% (Low fuel prices)** for scenarios without offshore wind (**Conservative Path: 4-8%**)
- If offshore wind and CO2 targets are enforced the cost premium of reaching net zero CO2 can be as high as **21%**
- Although **fuel price trajectories** can significantly **influence the total electricity prices**, this **effect can be minimized** if technologies follow Moderate or Advanced paths of development

Looking at Stochastic Optimization in Capacity Expansion Models – The Risk of Hurricanes

Hurricane Damage to Power Systems

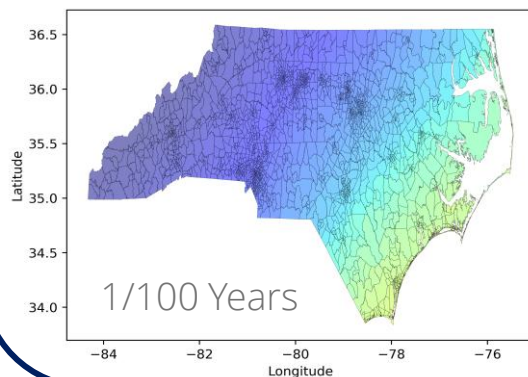
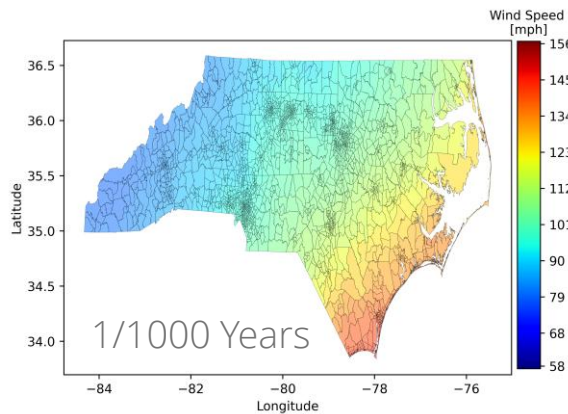
- Hurricane Irma hit Puerto Rico in 2017, followed by Hurricane Maria just two weeks later, leaving substantial damage to the power system infrastructure, which took more than 11 months to fully restore; following this event, FEMA approved a USD 9.5 billion plan aimed at reconstructing Puerto Rico's power system
- The estimated cost to North Carolina customers for the storm repairs made during 2018-2019 due to Hurricanes Florence, and Dorian account for more than US\$1 billion (Duke Energy 2021)
- Hurricane Irma (2017) is estimated to cost US\$1.3 billion to Florida Power & Light customers (FPL 2018). Hurricane Ian (2022) US\$1.1 billion (NextEra 2022)



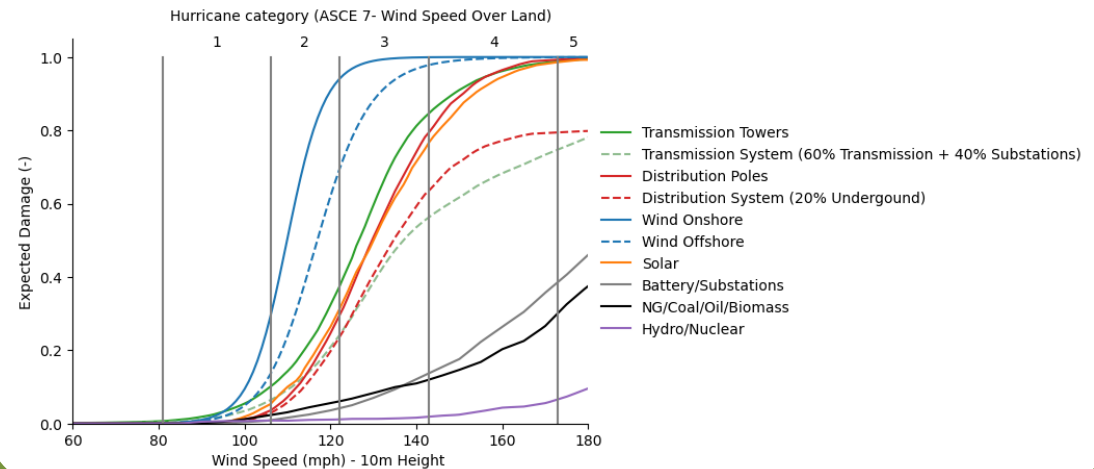
How to Model Hurricane Damage?

Extreme Wind Speeds From FEMA HAZUS Model (peak gust at 10m).

Different Recurrences:
20, 50, 100, 200, 500, 1000 yrs.



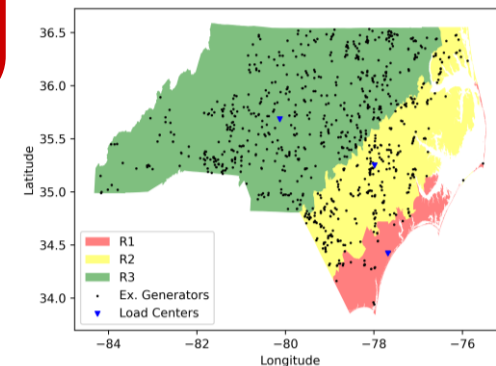
Fragility Curves



DAMAGE=Frangility (Wind Statistics)

DAMAGE for each technology type for
each county in the region of interest
and for different hurricane conditions

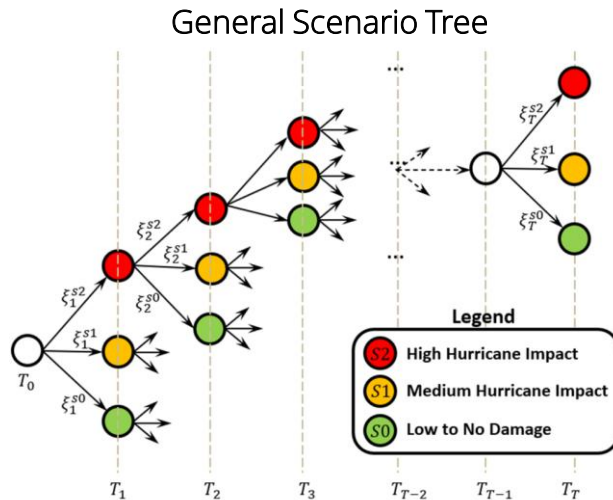
Regionalization Adopted



The Stochastic Model

Objective Function: minimizes the **expected cost** to meet energy demand throughout the entire time horizon.

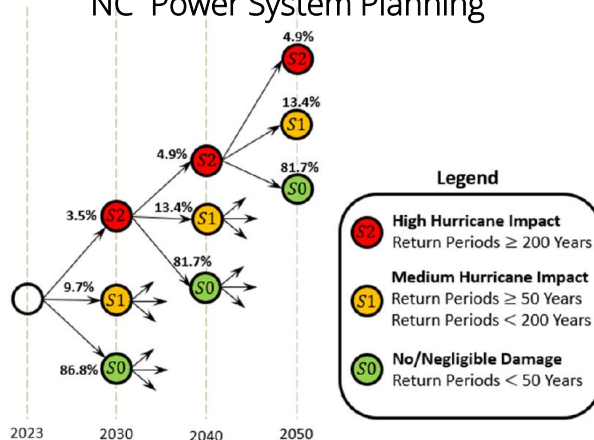
$$\min_{(x_0, y_0) \in \theta_0} \left\{ C_0(x_0, y_0) + \mathbb{E}_{\xi_1} \left[\min_{(x_1, y_1) \in \theta_1(x_0, \xi_1)} \left\{ C_1(x_1, y_1, \xi_1) + \dots \right. \right. \right. \\ \left. \left. \left. + \mathbb{E}_{(\xi_T | \xi_1, \dots, \xi_{T-1})} \left[\min_{(x_T, y_T) \in \theta_T(x_{T-1}, \xi_T)} C_T(x_T, y_T, \xi_T) \right] \right\} \right] \right\}$$



In each Scenario, Damage to Infrastructure is Compute as The Expected Damage, Conditioned on the Hurricane Intensities (Remember This is Technology and Region Dependent)

After the realization of a hurricane scenario, the existing **capacity of any technology is reduced proportionally to its expected damage**

Scenario Tree Used for The NC Power System Planning



Exponential Growth of Scenarios: Total # Scenarios = 3^n

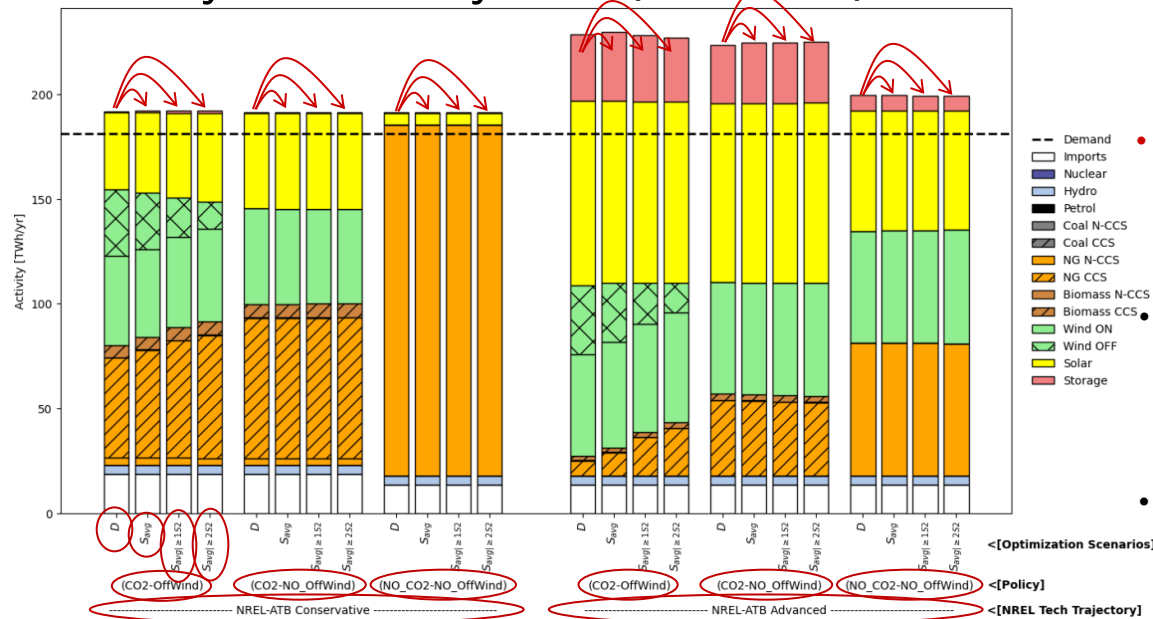
Deterministic vs. Stochastic



Deterministic: Assumes the Future Is Known (One Possible Realization- e.g., no Hurricanes)

Stochastic: Represents The Future as a Set of Possible Realizations (Scenarios) With Their Respective Probabilities

System Activity 2050 (R1, R2, R3)



- The system activity does not change significantly for scenarios without offshore wind when comparing the deterministic and the stochastic solutions

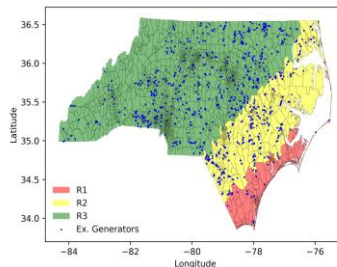
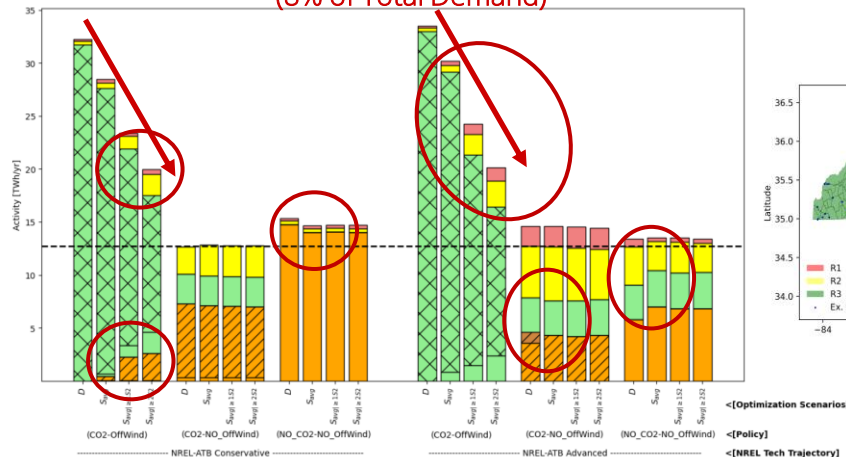
- For scenarios with offshore wind, changes in the model solution are more significant and involve (among other things) an increase in NG and solar energy participation.

When looking at the total system cost our results only shows very small increases from the deterministic solution without hurricanes to the stochastic solution with hurricanes.

- The largest difference in cost happens for the NREL-ATB Conservative technology simulations with CO2 and offshore wind targets, where an 0.8% increase can be observed between D and S_{avg} and 3.8% between D and $S_{avg} \geq 252$

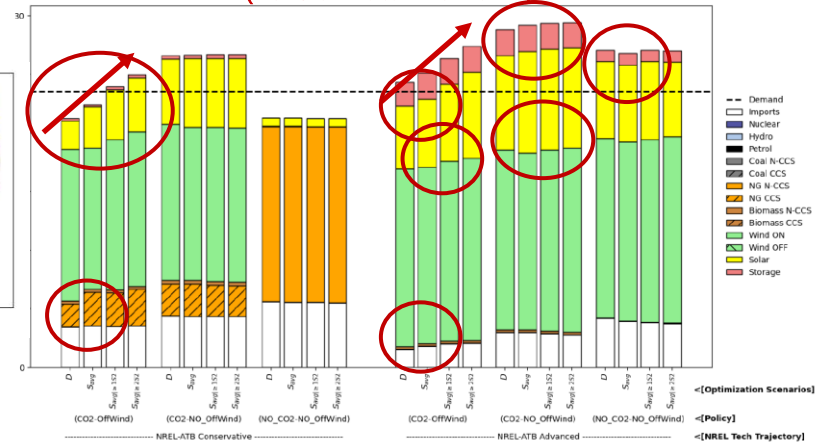
Activity 2050 Region R1

(8% of Total Demand)



Activity 2050 Region R2

(13% of Total Demand)

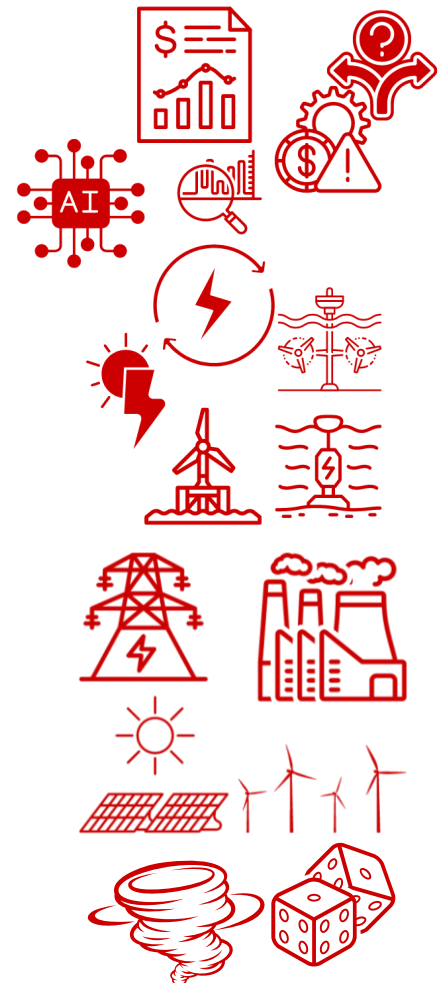


Conclusion

In the face of the **accelerated growth of renewable energy technologies**, predictive and prescriptive analytics will play a critical role in **reducing economic risk and maintaining the stability and reliability of our electric power sector**:

This dissertation contributed to this area by:

- Developing advanced predictive and prescriptive models to **enhance** renewable energy system **planning and operation**
- Developing innovative **portfolio optimization models**, helping to understand the opportunities for offshore renewable energy integration in North Carolina
- Developing and assessing **scenario generation** models for offshore renewable energy technologies, helping to improve uncertainty quantification in energy system analysis
- Investigating the effectiveness of Multi-Layer Perceptron Networks in **streamflow forecasting for hydropower** plants in Brazil, comparing its performance with models used by the Brazilian ISO
- Analyzing the **North Carolina energy sector's strategic responses** to policy changes, technology advancements, and natural disasters through 2050



Paper Contributions (2019-2024)

Journals (Approved and Under Review):

1. **de Faria, V. A.**, de Queiroz, A. R., Lima, L., Lima, J., and da Silva, B. (2021). **An assessment of multi-layer perceptron networks for streamflow forecasting in large-scale interconnected hydrosystems.** *International Journal of Environmental Science and Technology*.
2. **de Faria, V. A.**, de Queiroz, A. R., and DeCarolis, J. F. (2022). **Optimizing offshore renewable portfolios under resource variability.** *Applied Energy*.
3. **de Faria, V. A.**, de Queiroz, A. R., and DeCarolis, J. F. (2023). **Scenario generation and risk-averse stochastic portfolio optimization applied to offshore renewable energy technologies.** *Energy*.
4. Morais, L.B.S., Aquila, G., **de Faria, V. A.**, Lima, L. M. M., Lima, J. W. M., & de Queiroz, A. R. (2023). **Short-term load forecasting using neural networks and global climate models: An application to a large-scale electrical power system.** *Applied Energy*.
5. **de Faria, V. A.**, Jamaledin, N., de Queiroz, A. R., and Gabr, M. (2024). **Assessing the Risk of Hurricane Damage to Marine Hydrokinetic Devices.** *Applied Ocean Research* (under review).
6. **de Faria, V. A.**, de Queiroz, A. R., Sankarasubramanian, A., Ranjithan, R. (2024). **Route Optimization Under Flood Conditions.** *International Journal of Disaster Risk Reduction* (under review).
7. Nicoli, M., **de Faria, V.A.**, Savoldi, L., de Queiroz, A.R (2024). **Modeling Energy Storage in Long-Term Capacity Expansion Energy Planning: An Analysis of the Italian System.** *Journal of Energy Storage* (under review)
8. Nicoli, M., Colucci, G., **de Faria, V.A.**, Lerede, D., de Queiroz, A.R., Savoldi, L. (2024). **Evaluation of marginal and average sectorial GHGs emissions abatement costs using an open-source energy system optimization model: the Italian case study.** *Applied Energy* (under review).
9. **de Faria, V. A.**, Nicoli, M., de Queiroz, A. R. (2024). **The North Carolina Capacity Expansion Planning Under Alternative Government Policies Technology and Fuel Price Trajectories.** *Applied Energy* (in preparation).
10. **de Faria, V. A.**, Nicoli, M., Jamaledin, Gabr, M., N. de Queiroz, A. R. (2024). **Capacity Expansion Planning Under The Risk of Hurricanes - A Look Into the North Carolina Power System.** *Applied Energy* (in preparation).



Thank you!

Thank you



OPERATIONS RESEARCH
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Department of Civil, Construction, and Environmental
Engineering

Any Questions?

